SUMMER INTERNSHIP REPORT

Weight and Cost Reduction of Electrode Mast of the Ladle Heating Furnace

Submitted by

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Organization

SMS India Pvt. Ltd.

Metallurgy Department

Duration: 25 May 2026 – 3 July 2026 (6 Weeks)

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1. Internship Overview

Organization	SMS India Pvt. Ltd.
Department	Metallurgy Department
Duration	25 May 2026 – 3 July 2026 (6 Weeks)
Project Title	Weight and Cost Reduction of Electrode Mast of the Ladle Heating Furnace
Student	Ranveer Goyal – IIT Ropar (B.Tech. Mechanical Engineering)
Software Used	SolidWorks, ANSYS Mechanical 2019 Student

2. Introduction

2.1 About SMS India Pvt. Ltd.

SMS India Pvt. Ltd. is the Indian subsidiary of SMS group GmbH, a world-leading supplier of plant and machinery for the steel and non-ferrous metals industry. The Metallurgy Department focuses on primary and secondary steelmaking equipment including ladle furnaces and vacuum degassers. SMS group has supplied over 150 single-station and 40 twin-station ladle furnaces globally, with Indian installations at Bokaro, JSPL Angul, Tata Steel, JSW, SAIL, and AM/NS among others.

2.2 Project Objective

The objective of this project was to perform static structural FEA of the existing electrode mast and arm assemblies of a Ladle Heating Furnace (LHF), and to redesign selected mast cross-sections to achieve weight and material cost reduction while maintaining structural integrity under all operating loads.

- Study the LHF equipment and the steelmaking process.
- Perform manual force and moment calculations for the electrode mast-arm assembly.
- Rebuild simplified 3D models in SolidWorks suitable for FEA.
- Conduct static structural analysis in ANSYS Mechanical 2019 Student for both Ø508 mm and Ø457 mm electrode variants (side arm E-ARM#1 and middle arm E-ARM#2).
- Propose and validate redesigned mast cross-sections with reduced weight and cost.

3. Steelmaking Process Overview

As background for the LHF equipment studied in this project, a brief overview of the steelmaking route is given below.

Iron ore is processed through a sinter plant and reduced in a Blast Furnace (or via DRI) to produce liquid hot metal. This is converted to steel in a primary furnace — a Basic Oxygen Furnace (BOF), Electric Arc Furnace (EAF), or Conpro Furnace — depending on the plant configuration. The liquid steel is then tapped into a ladle and moved to the Ladle Heating Furnace (LHF) for secondary refining, where arc heating from graphite electrodes maintains the precise temperature required for casting, while alloying, desulphurization, and inert-gas stirring adjust the final chemistry. For high-grade steels, the ladle may additionally pass through a degassing unit (RH or VD/VOD) to remove dissolved gases. Finally, the refined steel is sent to the Continuous Casting Machine (CCM) and solidified into billets, blooms, slabs, or rounds for downstream rolling.

The LHF — the focus of this project — therefore functions as a critical buffer and refining station between primary steelmaking and casting, and its electrode mast-arm assembly must reliably withstand continuous mechanical and thermal loads during operation.

4. Ladle Heating Furnace – Equipment Description

4.1 LHF Types

Single-station LHFs have one treatment position and are compact and energy-efficient (30–350 t). Twin-station LHFs have two treatment positions sharing a common power supply, offering greater operational flexibility and are preferred for superior-quality steel grades (120–350 t).

4.2 Key Components

- Water-Cooled Roof: Dome-shaped or skirt-type, made of water-cooled pipes. Fitted with pneumatically-operated flap gates for sampling and temperature measurement, an alloy chute, wire feeding funnel, and fume elbow connecting to the fume extraction system.
- Electrode Regulation System: The SMS X-Pact® SynReg system regulates electrode position via a primary impedance/current control loop and a secondary dynamic adjustment loop, minimizing electrode consumption and energy use.
- Electrode Arms and Mast: The structural assembly that holds and positions the graphite electrodes (detailed in Section 5).
- High Current System: Flexible water-cooled cables carry the arc current from the transformer to the electrode arms.
- Argon Top Lance / Bottom Purging: For emergency stirring and inert gas homogenization through porous plugs.
- Wire Feeding Station: Feeds alloy or treatment wires into the ladle.
- Automatic Temperature & Sampling Lance: For automated steel temperature measurement and sampling.
- Fume Extraction System: Captures and cleans furnace off-gases.

5. Electrode Mast-Arm System Description

5.1 System Overview

The electrode mast-arm assembly is the structural backbone of the LHF. It supports and positions the three graphite electrodes (one per phase of the three-phase arc system) over the ladle. The system consists of one mast per electrode phase with a cantilevered arm. The mast translates vertically via a hydraulic cylinder to raise and lower the electrode.

5.2 Major Components

- Electrode Mast: The primary vertical structural member. It is a water-cooled hollow rectangular box section, with mast height varying from 6.8 m to 9.5 m depending on the configuration. Including the hydraulic cylinder stroke, the total assembly height can reach up to ~15 m. Internal water chambers (left, middle, and right) cool the mast during operation.
- Electrode Arm (E-ARM#1 – Side Arm, E-ARM#2 – Middle Arm): Cantilevered arms extending horizontally from the mast, carrying the graphite electrode at their tip. The arms contain upper and lower water-cooling chambers. The middle arm (E-ARM#2) is centrally located; the side arms (E-ARM#1) are positioned on either side.
- Graphite Electrode: Conducts the arc current to the steel bath. Standard diameters analyzed are 508 mm and 457 mm. Subject to arc force (slag force) at the tip.
- Insulating Plate: Electrically isolates the electrode arm from the mast to prevent current leakage to the structure. Connected via bolted joints.
- Support Rollers – Moment-Resisting Pair: The mast is guided along a vertical rail by a total of 8 rollers (front and back, left and right, upper and lower). One pair of diametrically-opposed rollers bears against the guide rail and reacts the overturning bending moment generated by the cantilevered arm weight. The vertical spacing between this roller pair is 3100 mm for the Ø508 mm electrode configuration (both arms) and 2900 mm for the Ø457 mm electrode configuration (both arms). The upper roller pushes in one direction; the lower roller reacts in the opposite direction, forming a moment couple.
- Support Rollers – Thermal Gap Pair: The rollers on the opposite side of the moment-resisting pair maintain a small clearance gap between them and the mast rail. This gap accommodates thermal expansion of the mast during furnace operation and prevents them from generating unwanted constraint loads under normal conditions.
- Hydraulic Cylinder: Raises and lowers the entire mast-arm assembly for electrode positioning. The cylinder rod (bottom) is fixed to the gantry structure.
- Clamping Unit: Clamps the graphite electrode securely at the arm tip. Its weight (732 kg) acts at the top surface of the arm tip.
- Water Cooling Cables (W.C. Cables): Flexible water-cooled cables carrying high current. Their weight, taken as 1565 kg, acts at the rear side of the arm.
- Cooling Water Hoses: Supply and return hoses for mast cooling water.

5.3 Electrode Variants Analyzed

Electrode Dia.	Arm Type	Mast 1 (Original) X-section	Mast 2 (Redesigned) X-section
508 mm	E-ARM#2 (Middle Arm)	600 × 410 mm	550 × 350 mm + stiffener plate
508 mm	E-ARM#1 (Side Arm)	600 × 410 mm	550 × 350 mm + stiffener plate
457 mm	E-ARM#1 (Side Arm)	550 × 400 mm	Not redesigned (Mast 1 only)
457 mm	E-ARM#2 (Middle Arm)	550 × 400 mm	500 × 310 mm

For the Ø457 mm electrode configuration, the redesigned Mast 2 (500 × 310 mm) also incorporated plate thickness changes: the original 40 mm major-axis plate was increased to 42 mm, while the 25 mm minor-axis plate was reduced to 20 mm. For the Ø508 mm electrode configuration, the major-axis plate thickness (42 mm) was kept unchanged, while the minor-axis plate thickness was reduced from 25 mm to 20 mm.

5.4 Materials Used

Mast and Arm Structure (Structural Steel S355):

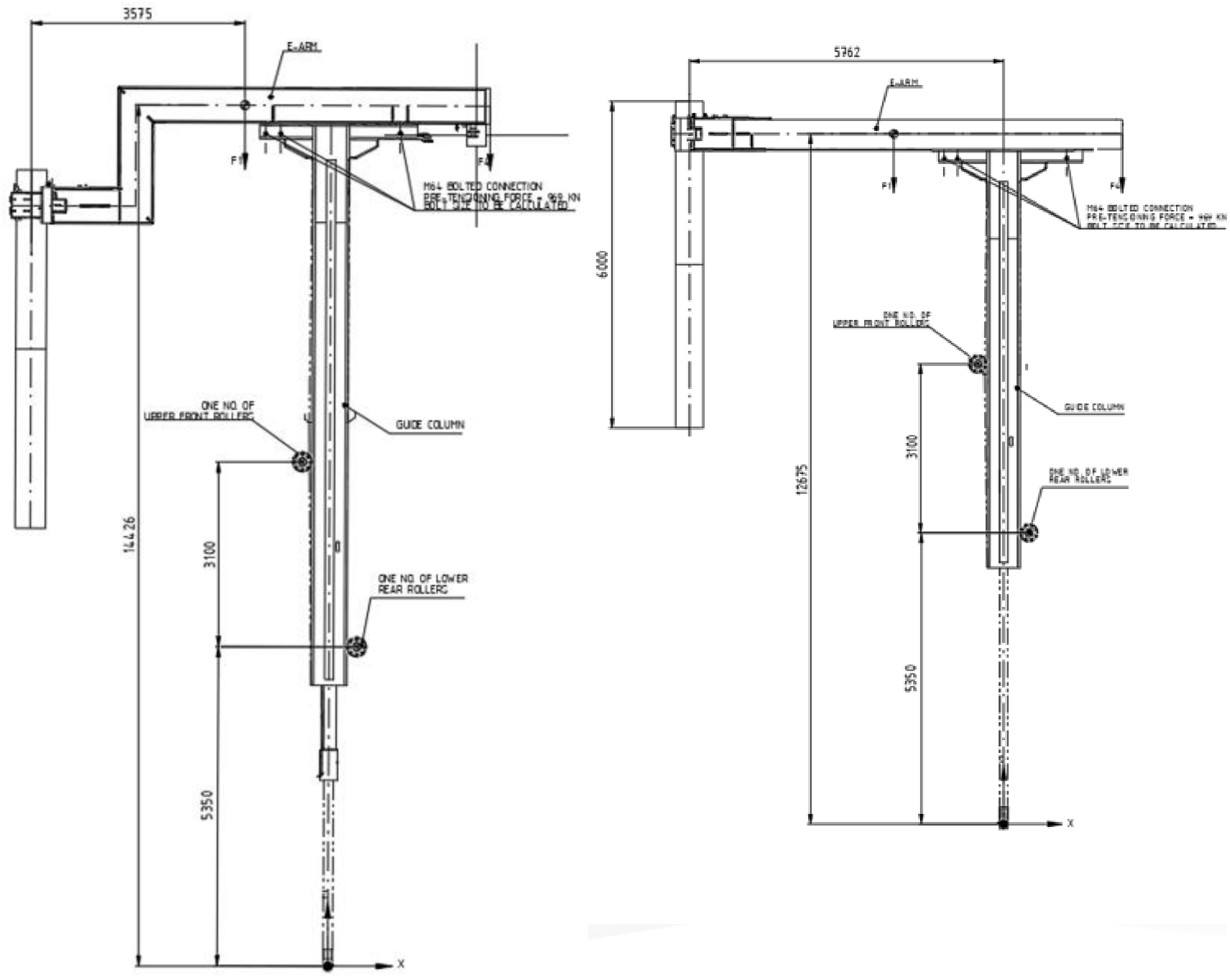
Property	Value	Unit
Material Grade	S355	–
Yield Strength f_y	355	MPa
Ultimate Strength f_u	510	MPa
Elastic Modulus E	210,000	MPa
Shear Modulus G	81,000	MPa
Density ρ	7,850	kg/m^3
Poisson's Ratio ν	0.3	–
Thermal Expansion α	1.2×10^{-5}	/°C
Temperature Rise (operating)	110	°C
Thermal Stress $\sigma_{thermal}$	6.3	MPa

Insulating Plate (EP GC 203 – epoxy glass-cloth laminate):

Property	Value	Unit
Material Grade	EP GC 203	–
Density ρ	2,470	kg/m^3
Young's Modulus E	22	GPa
Poisson's Ratio ν	0.2	–

Electrode (Graphite):

Property	Value	Unit
Material	Graphite	–
Density ρ	1,800	kg/m³
Young's Modulus E	10	GPa
Poisson's Ratio ν	0.2	–



1. Middle and side sketches

6. Manual Force and Moment Calculations

Before proceeding with FEA, manual calculations were performed to estimate the reaction forces at the support rollers and the load on the hydraulic cylinder base. These results were used to verify the FEA model.

6.1 Methodology

The entire electrode mast-arm assembly was treated as a free body. The total bending moment about the support roller pair was found using the centre of gravity (CG) and weight of each component. Roller reactions were then calculated using moment equilibrium:

$$Ra = M / A \quad \text{and} \quad Rb = -Ra$$

where A = roller vertical spacing (3100 mm for the Ø508 mm electrode, both arms; 2900 mm for the Ø457 mm electrode, both arms)

The total vertical (axial) load on the hydraulic cylinder base equals the sum of all component weights, including the FOS-adjusted effective densities described in Section 7.3.

Cross-product mechanics were used to resolve the moments generated by each load (self-weight, water, dust, cables, clamping unit, slag) about the roller axis, producing the bending moment and roller reaction forces used to cross-check the corresponding ANSYS results in Section 9.

7. 3D Model Preparation in SolidWorks

7.1 Original Models

The original CAD assemblies of the electrode mast-arm system were provided in SolidWorks/STEP format. Each assembly included the mast, arm, electrode, insulating plate, support rollers, hydraulic cylinder, and numerous auxiliary components: pipes, hose connectors, clamps, cable brackets, and instrumentation fittings.

7.2 Model Simplification Strategy

Small auxiliary components significantly increase mesh element count and computation time without meaningfully affecting the global stress or deformation results. The following simplification approach was adopted:

- Removed: Pipes, hose fittings, small clamps, cable brackets, and miscellaneous hardware.
- Retained: Mast body (with internal water chambers), electrode arm, graphite electrode, insulating plate, support rollers, and hydraulic cylinder.
- Hollow internal volumes of water chambers in the mast (left/middle/right) and arm (upper/lower) were preserved intact for accurate water volume and pressure loading.
- Bodies were merged where necessary and internal geometry verified using SolidWorks Section View and Mass Properties.

7.3 Effective Density Compensation

To account for the mass of removed components without re-adding geometric complexity, the effective density of the mast and arm bodies was adjusted:

$$\rho_{eff} = (\text{Real Total Mass of Assembly}) / (\text{Simplified Model Volume})$$

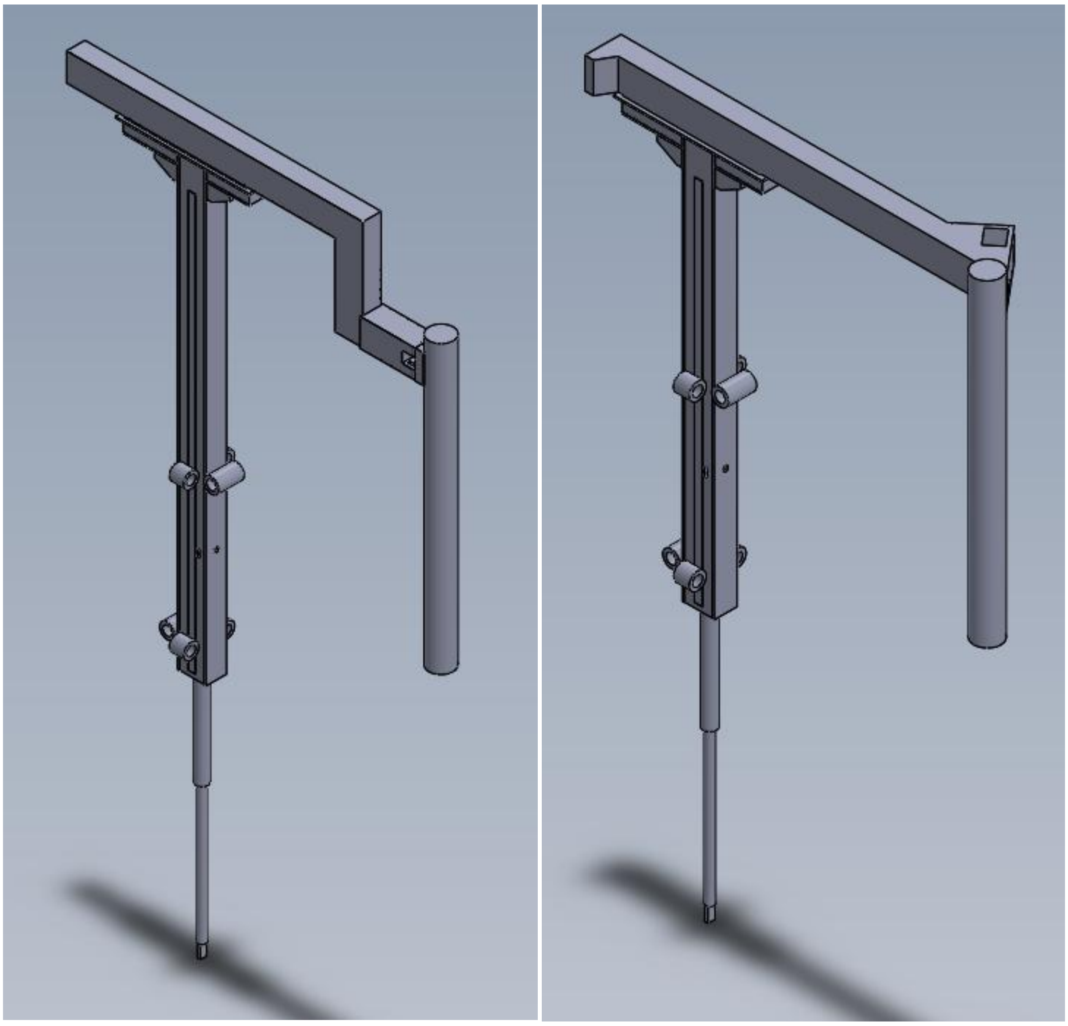
An additional Factor of Safety (FOS) of 1.3 was then applied by multiplying the effective density by 1.3, to conservatively account for thermal stresses and operational dynamic loads (vibration, impact from sudden stopping):

$$\rho_{applied} = 1.3 \times \rho_{eff}$$

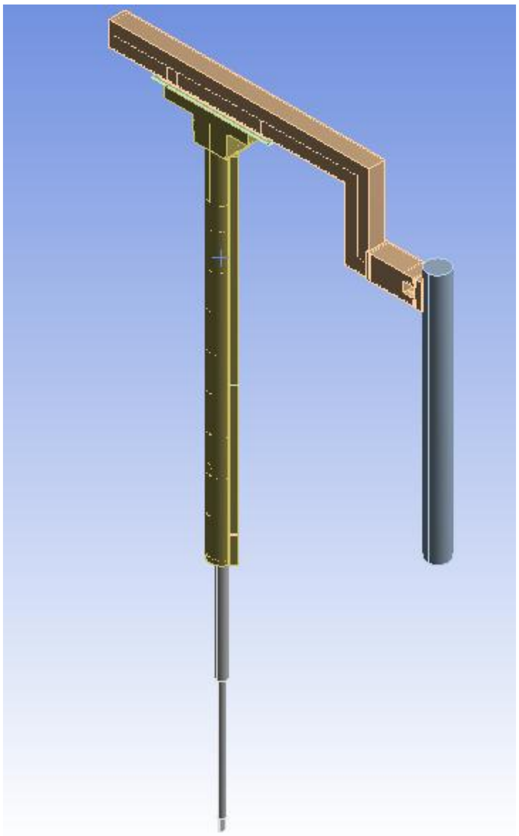
This same FOS of 1.3 was also applied to the water weight loads and other external forces (cables, clamping unit) to produce conservative loading. The dust load and slag force were applied without this FOS multiplier.

7.4 STEP File Export

Simplified assemblies were exported as STEP (.stp) files for import into ANSYS Mechanical 2019. STEP format ensures geometry fidelity and is fully compatible with ANSYS DesignModeler.



2. Middle and Side Arm Assemblies



3. Cylindrical Mast

8. Finite Element Analysis – ANSYS Mechanical 2019

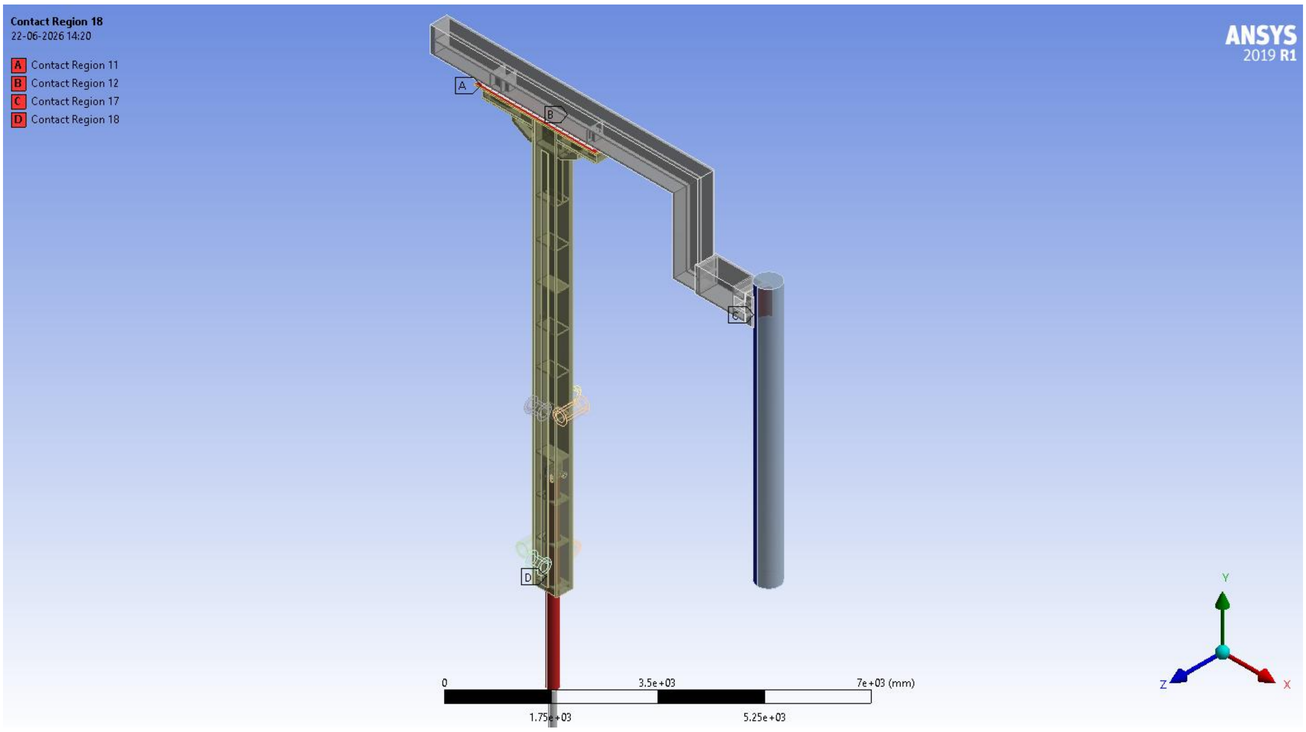
8.1 Material Properties

Material properties were entered in ANSYS Engineering Data for each part: structural steel (S355) for the mast, arm, and hydraulic cylinder with adjusted effective density (see Section 7.3); EP GC 203 epoxy glass-cloth laminate for the insulating plate; and graphite for the electrode (properties listed in Section 5.4).

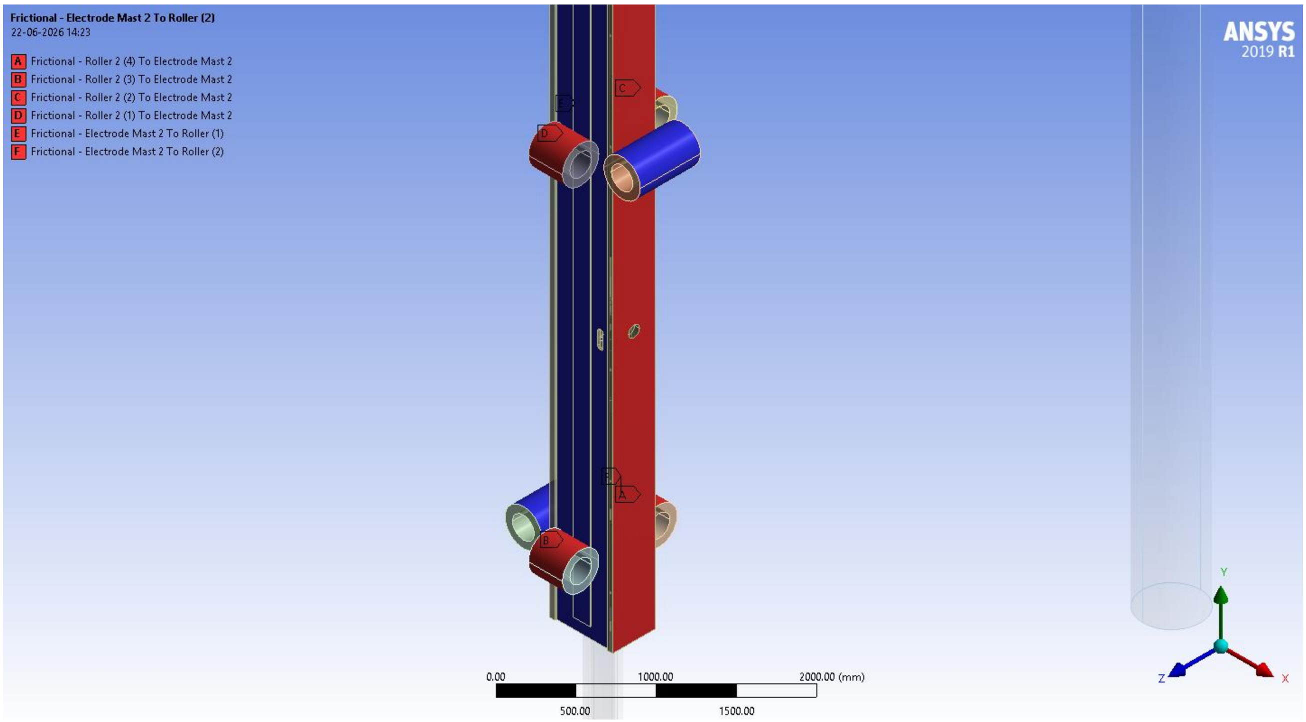
8.2 Contact Definitions

The following contact pairs were defined, based on the actual contact setup used for the side and middle arm models:

Contact Pair	Type	Remarks
Support Rollers – Mast (guide column)	Frictional (μ = 0.03)	Upper-front and lower-rear roller only; 6 mm strip for surface contact.
Mast (Guide Column) – Electrode Arm	Bonded	Full attachment
Mast – Insulation – Electrode Arm	Bonded	Three-body bonded joint
Mast – Hydraulic Cylinder	Bonded	Full attachment
Electrode – Electrode Arm	Bonded	Full attachment



4. Bonded Contacts



5. Frictional Contacts

8.3 Bolt Pretension Study

The insulating plate is connected to the mast and arm by bolted joints. Applying bolt pretension in ANSYS requires a multi-step solver with separate load steps, significantly increasing computation time. A comparative study was performed:

- Case A: Full bolt pretension on all bolts.
- Case B: Bolts replaced with bonded contacts (no pretension).

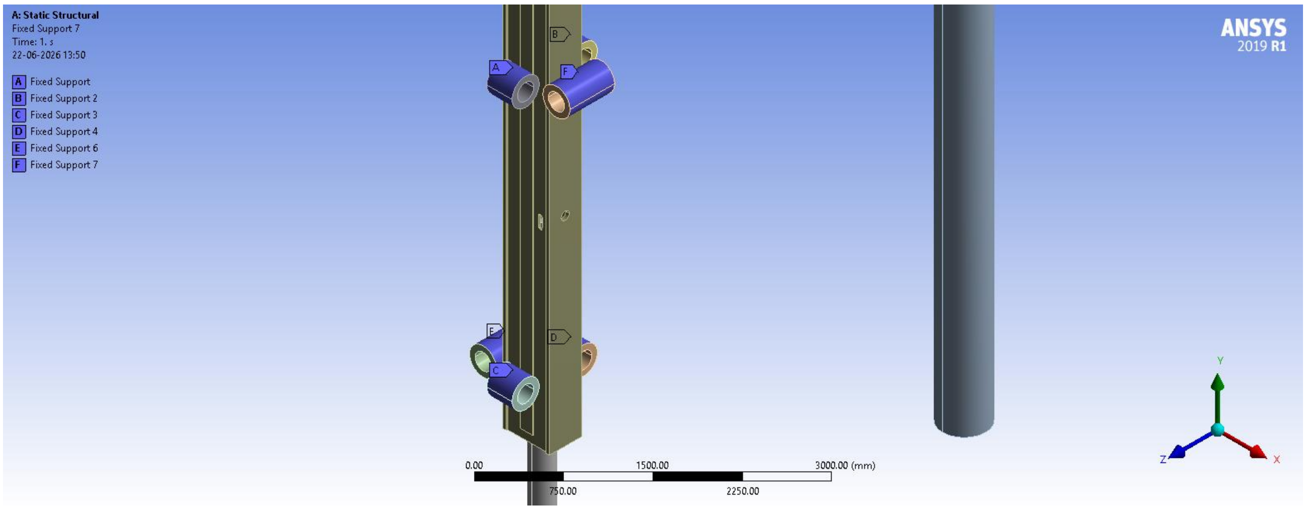
Findings:

- Total deformation difference: Less than 0.1 mm (negligible for this scale of assembly).
- Equivalent stress difference: Only local stress concentrations at bolt locations due to the pretension force; the global stress field was essentially identical.

Conclusion: Bonded contact is an adequate approximation for this global structural assessment. All subsequent analyses used bonded contacts, saving significant computation time.

8.4 Boundary Conditions

All support rollers (the moment-resisting line-contact pair as well as the remaining rollers) were given fixed support in every configuration analyzed. The hydraulic cylinder rod (bottom) was also fixed. The roller pair on the side opposite the moment-resisting pair maintains a clearance gap from the mast rail to accommodate thermal expansion (Section 5.2) and is therefore not engaged under normal operating loads.



6. Fixed Constraints on rollers

8.5 Mesh Settings

Global element size: 100 mm. Mesh sensitivity was assessed at 75 mm; peak deformation difference was only 0.1–0.2 mm, confirming that 100 mm is adequate for this large assembly (mast height 6.8–9.5 m, with the total assembly height — including hydraulic cylinder stroke — reaching up to ~15 m).

Component	Mesh Method	Reason
Electrode	Sweep	Cylindrical geometry
Support Rollers	Sweep	Cylindrical geometry
Mast, Arm, Insulating Plate, Hydraulic Cylinder	Tetrahedral Patch Conforming	Complex irregular geometry

8.6 Applied Loads

8.6.1 Standard Earth Gravity

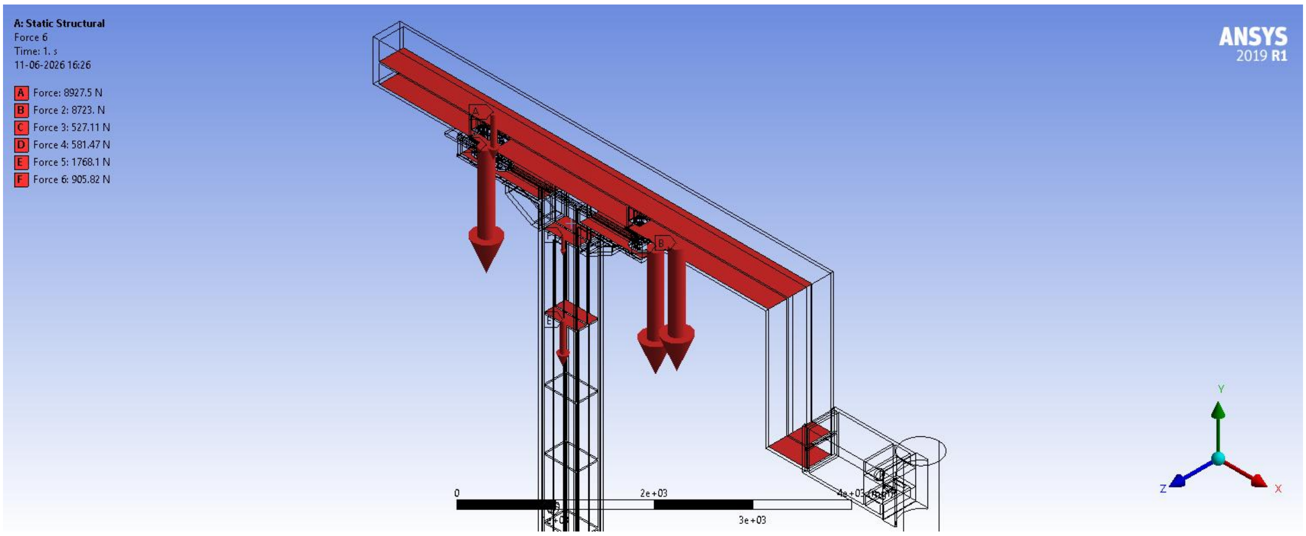
Gravity of 9.81 m/s² applied in the −Y direction to account for self-weight of all components.

8.6.2 Cooling Water Weight

The weight of static cooling water in each chamber was calculated using the internal volume measured in SolidWorks:

$$F_{water} = V_{chamber} \times \rho_{water} \times g$$

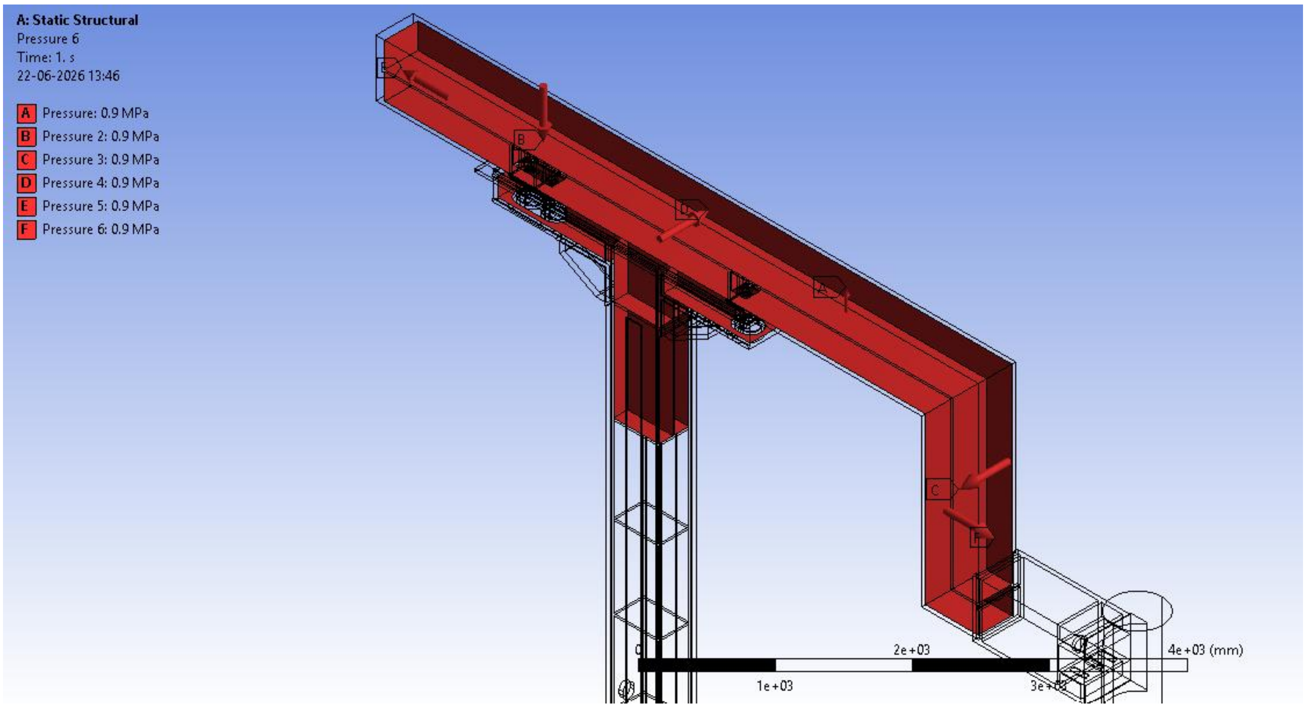
Applied separately for: Mast left chamber, Mast middle chamber, Mast right chamber, Arm upper chamber, Arm lower chamber. Water weight forces were multiplied by FOS = 1.3.



7. Weight of Water downwards

8.6.3 Dynamic Water Pressure

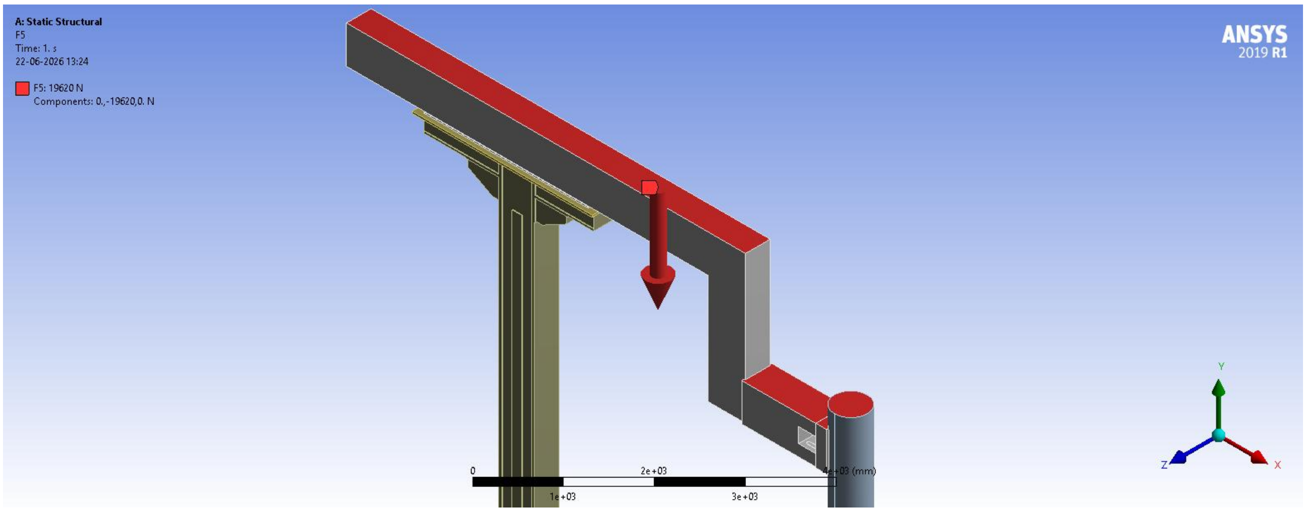
Since cooling water flows continuously at high pressure, a dynamic pressure of 0.9 MPa (9 bar) was applied uniformly as a surface pressure on the side surfaces of the lower and upper chambers, directed outward (normal to each surface). This represents the internal pressure acting to expand the chambers.



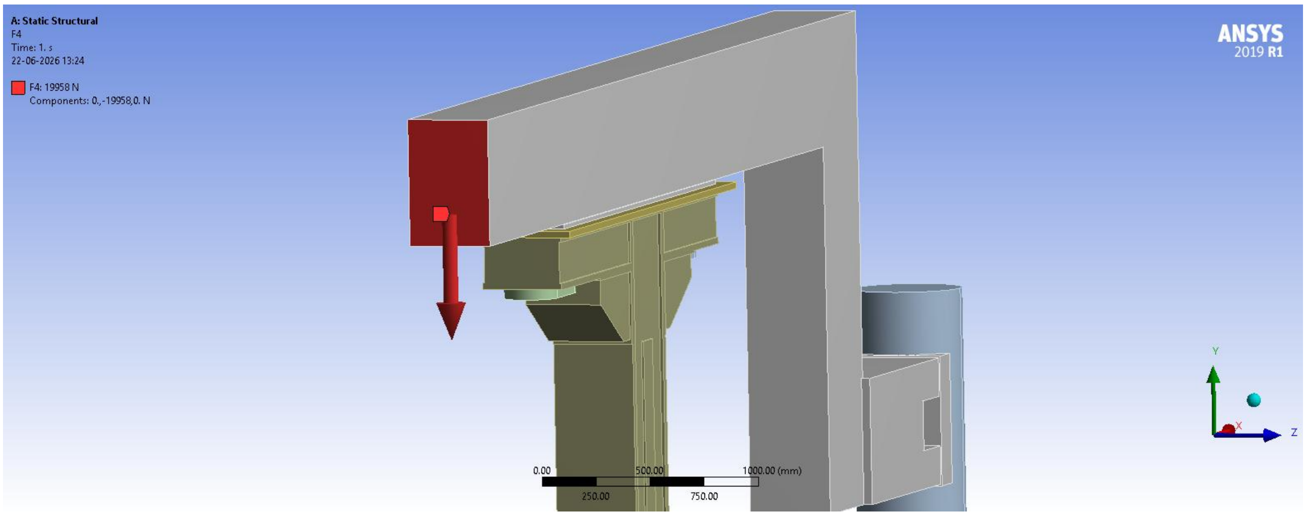
8. 9 Bar Pressure outwards on all walls with water

8.6.4 Additional External Loads

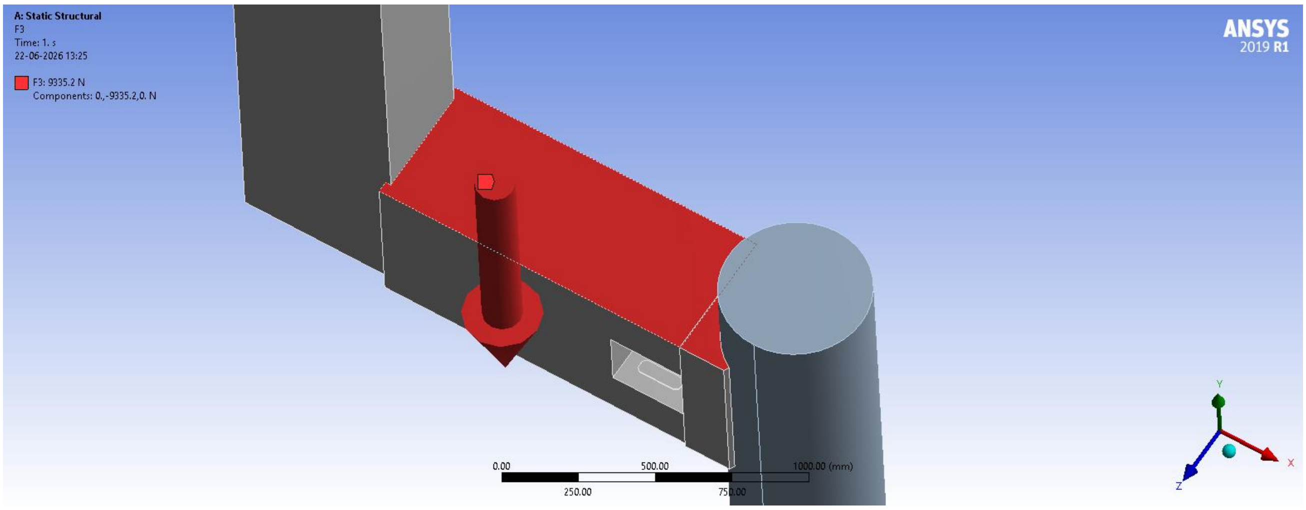
Load	Value	Applied Location	FOS Applied
Dust load	2000 kg	Top surfaces	No
Water-cooled cables weight	1565 kg	Rear face of arm	Yes (×1.3)
Electrode de-clamping unit weight	732 kg	Top surface, arm tip	Yes (×1.3)
Slag force (electrode tip)	509.68 kg	Side of electrode	No



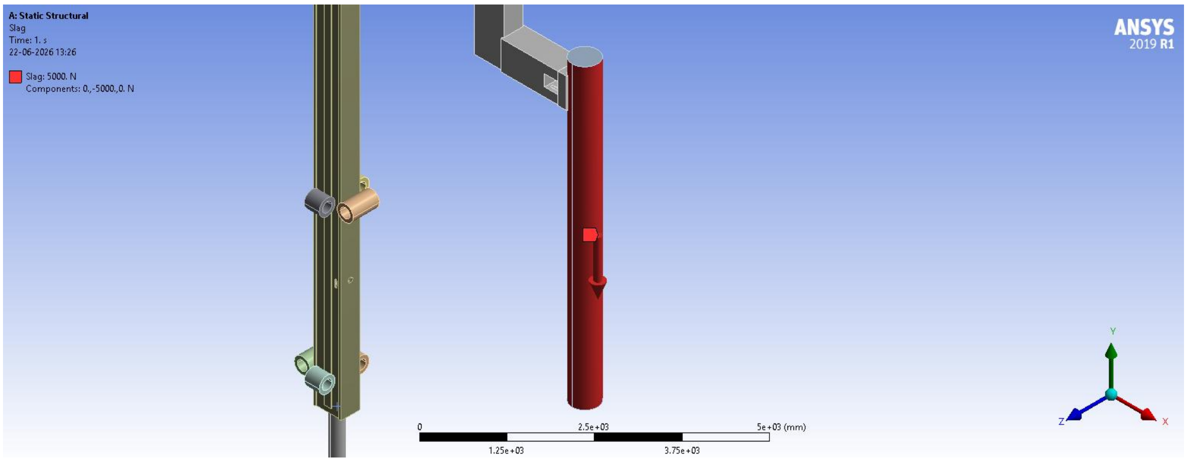
9. *Dust Load*



10. *Weight of water-cooled Cables*



11. *Weight of Clamping Unit*



12. Slag Load

9. FEA Results

For each configuration, the following results were extracted: Equivalent (Von Mises) Stress, Total Deformation, Directional Deformation (X, Y, and Z where recorded), and Reaction Forces at the support rollers and hydraulic cylinder base. Directional deformation values are reported as magnitudes — the sign in the raw ANSYS output only indicates direction along the respective axis. All forces below include FOS = 1.3 already applied where applicable. Image placeholders are left below each result for the corresponding ANSYS plots. All values obtained are within the allowable deformation limits for the assembly.

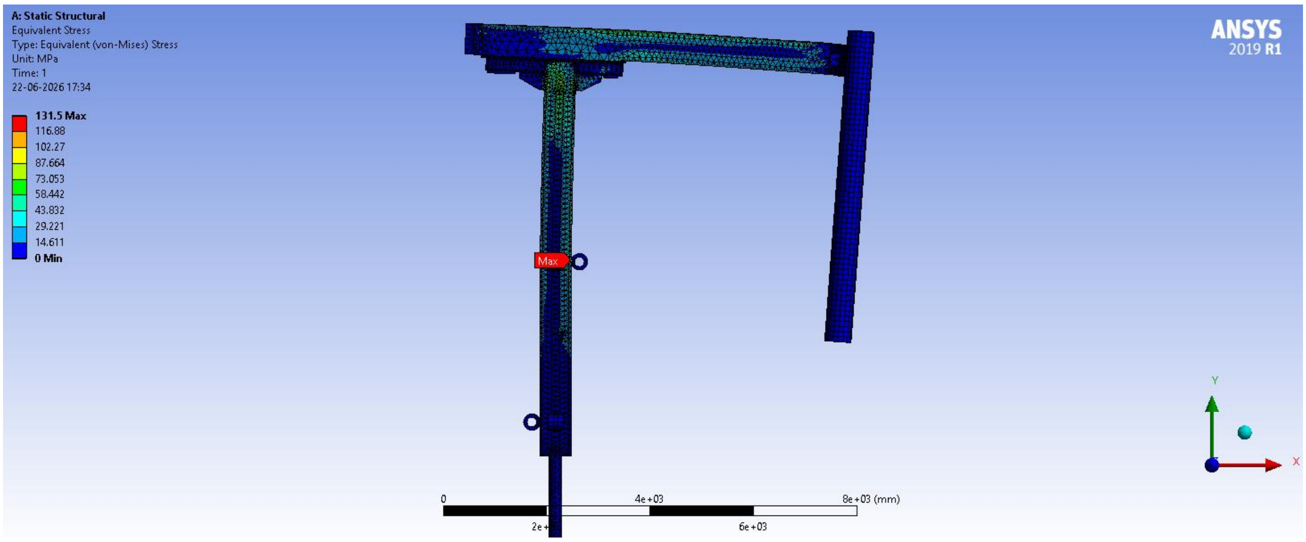
9.1 E-ARM#1 (Side Arm) – Ø508 mm Electrode

Electrode length: 6000 mm. Note: weight of the online electrode jointer is not considered. Roller vertical spacing: 3100 mm.

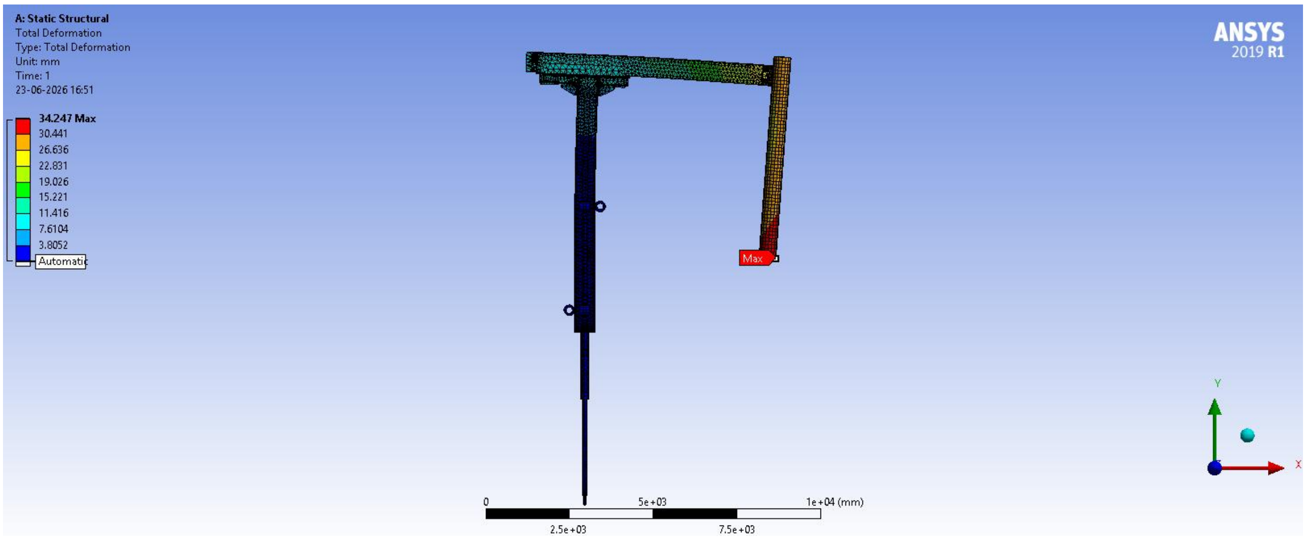
9.1.1 Mast 1 (Original: 600 mm × 410 mm)

Loads: Upper chamber water 8243 N, Lower chamber water 7826.9 N, Mast water (W3, W4, W5) 698.62 N / 614.07 N / 3521 N, Dust load (no FOS), Water-cooled cables + water 19958 N, Slag on electrode (no FOS), Electrode de-clamping unit weight 9335.2 N, Pressure 0.9 MPa (9 bar) applied on the side surfaces of the lower and upper chambers.

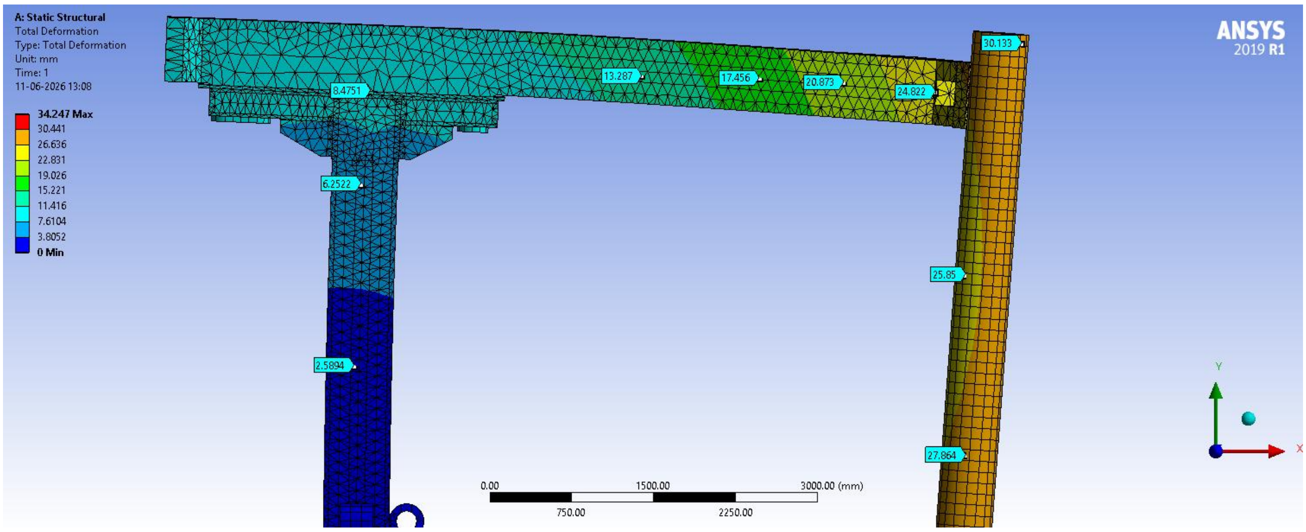
Equivalent (Von Mises) Stress – Max (in mast)	131.5 MPa
Total Deformation – Max (at electrode tip)	34.247 mm
Deformation – Electrode Arm (range)	12 – 25 mm (max 27 mm at clamping area)
Deformation – Mast	8.5 mm
Directional Deformation X – Max	19.823 mm
Directional Deformation Y – Max	27.731 mm
Directional Deformation Z – Max	4.755 mm
Reaction Force – Upper Roller (–ve x-direction)	127,900 N
Reaction Force – Lower Roller (+ve x-direction)	127,750 N
Reaction Force – Cylinder Rod	192.48 kN



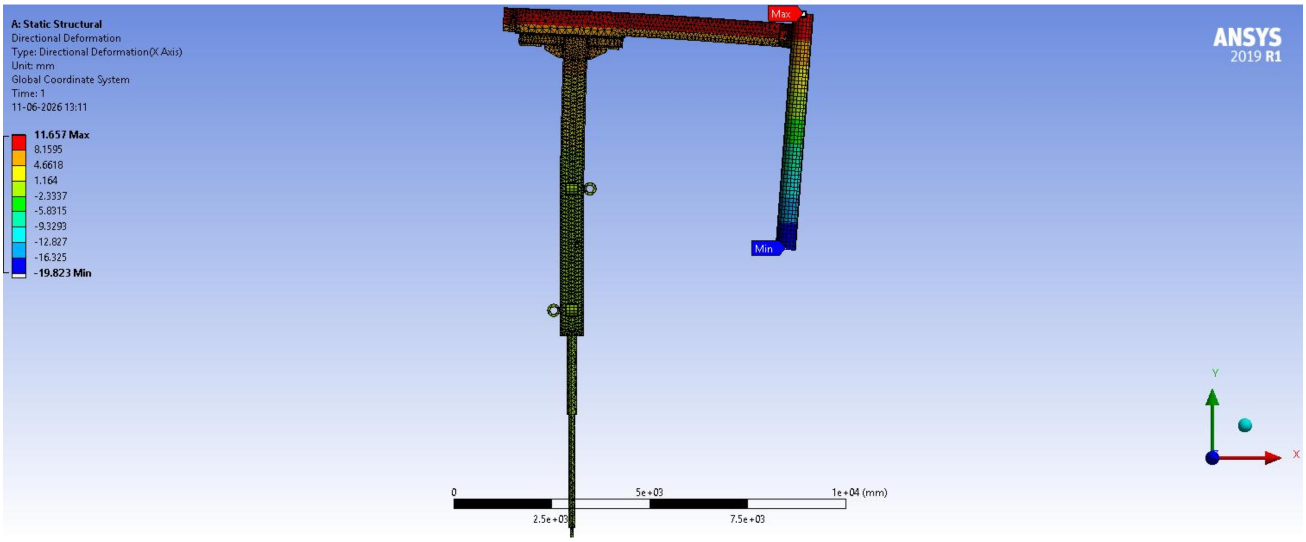
13. Equivalent Stress



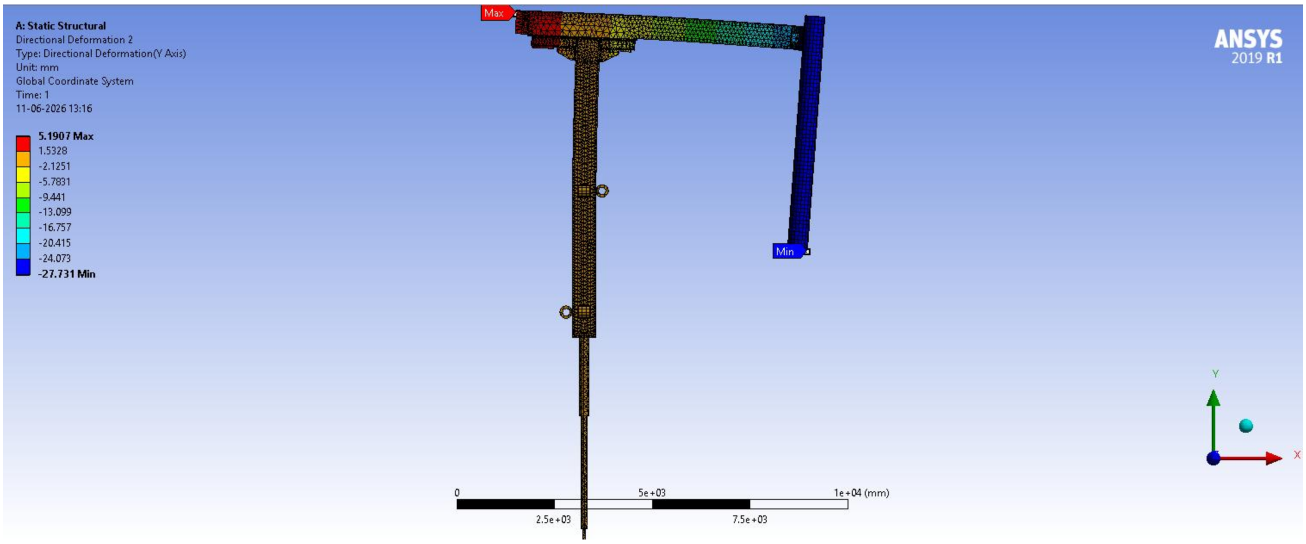
14. Total Deformation



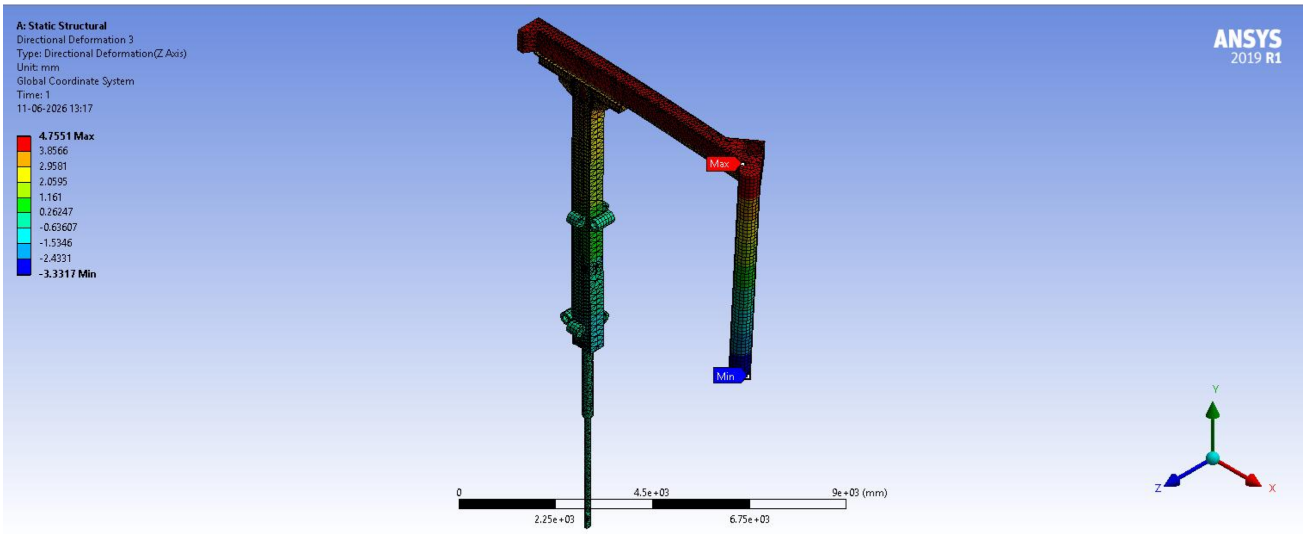
15. Deformation probes



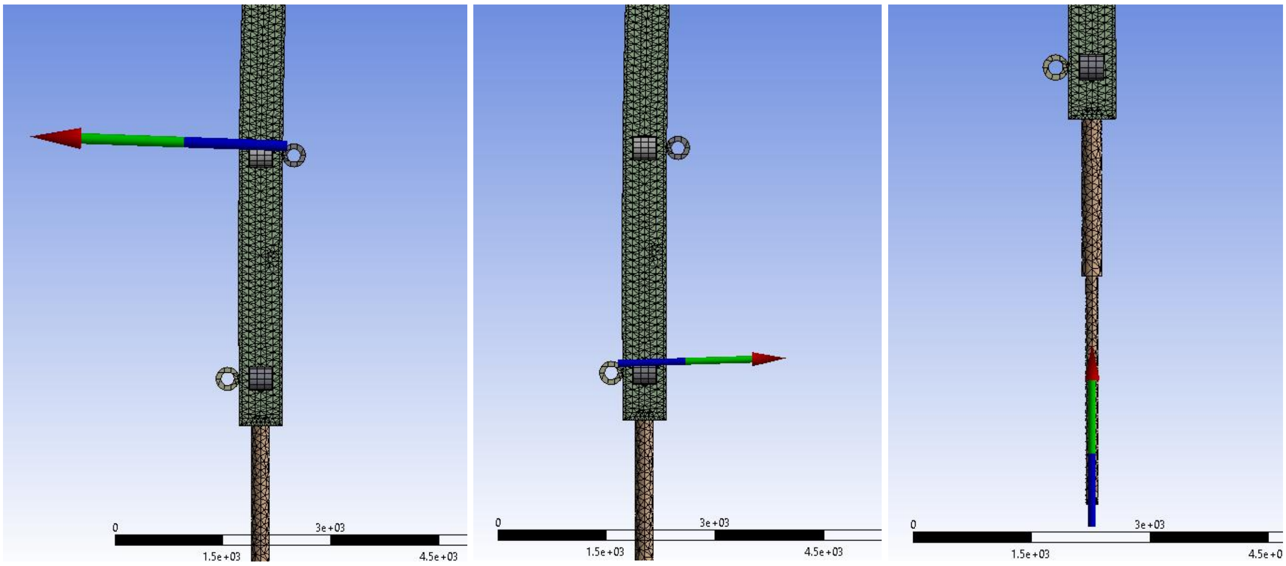
16. X-axis Deformation



17. Y-axis Deformation



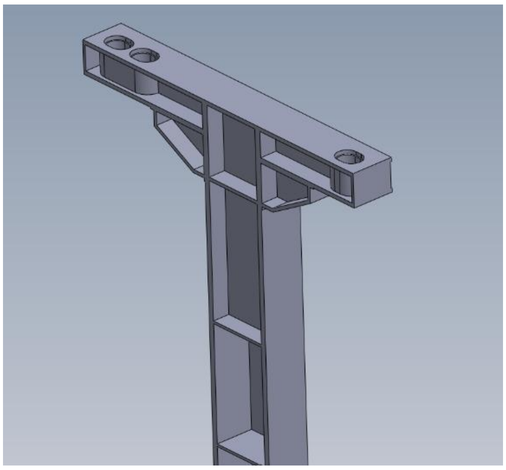
18. Z-axis Deformation



19. Reaction Forces on the Rollers and Hydraulic Cylinder

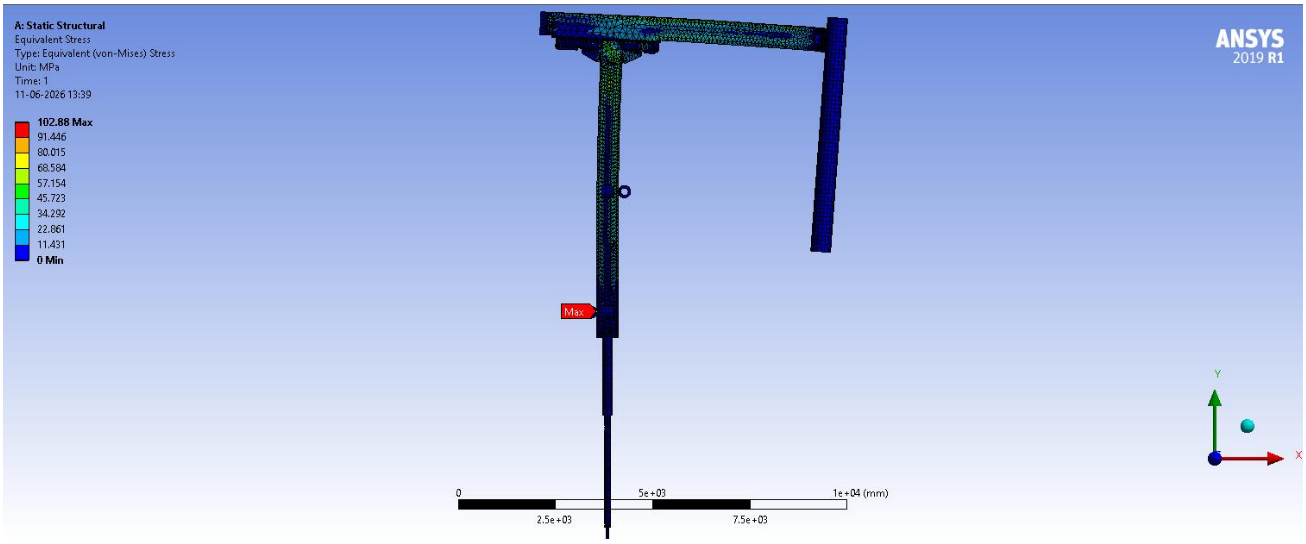
9.1.2 Mast 2 (Redesigned: 550 mm × 350 mm, plate added at mast top)

Cross-section altered from 600 × 410 mm to 550 × 350 mm; a plate was added to the top of the mast to counter stress. All constraints, boundary conditions, and electrode-arm load conditions are unchanged from Mast 1. Mast water loads updated: W3 (699.62 N → 581.47 N), W4 (614.07 N → 527.11 N), W5 (3521 N → 1768.1 N); the added top plate exerts an additional 905.82 N due to the water column.

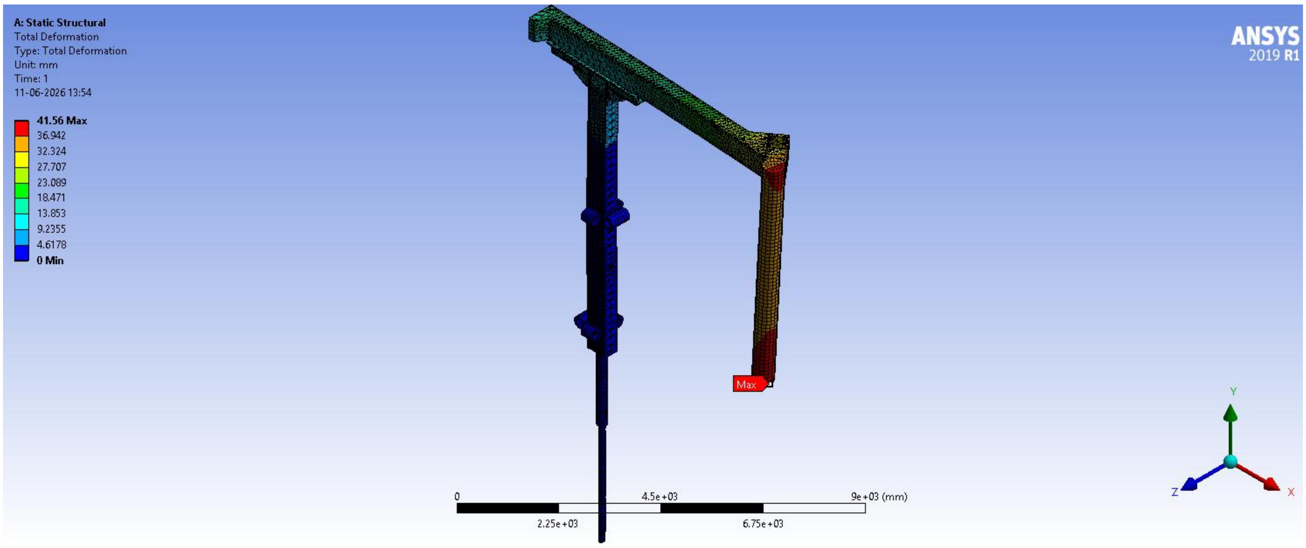


20. Added an extra plate in the mast

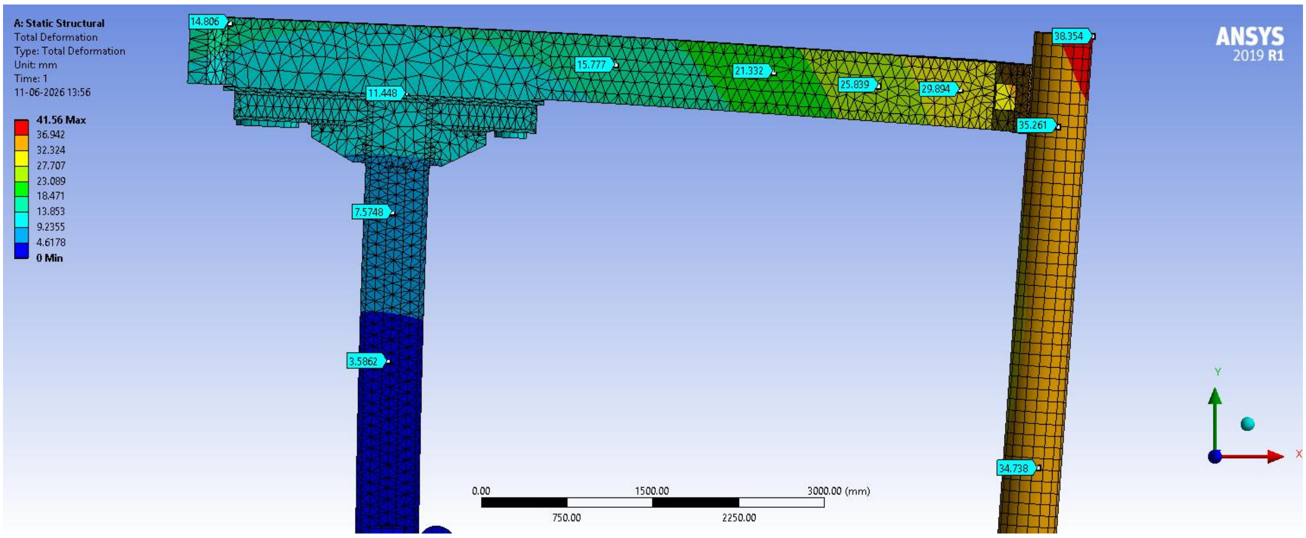
Equivalent (Von Mises) Stress – Max (in mast)	102.88 MPa
Total Deformation – Max (at electrode tip)	41.56 mm
Deformation – Electrode Arm (range)	15 – 30 mm (max 35 mm at gripping point)
Directional Deformation X – Max	22.64 mm
Directional Deformation Y – Max	34.66 mm
Reaction Force – Lower Roller (+ve x-direction)	127.8 kN
Reaction Force – Upper Roller (–ve x-direction)	128 kN
Reaction Force – Cylinder Rod	178.84 kN



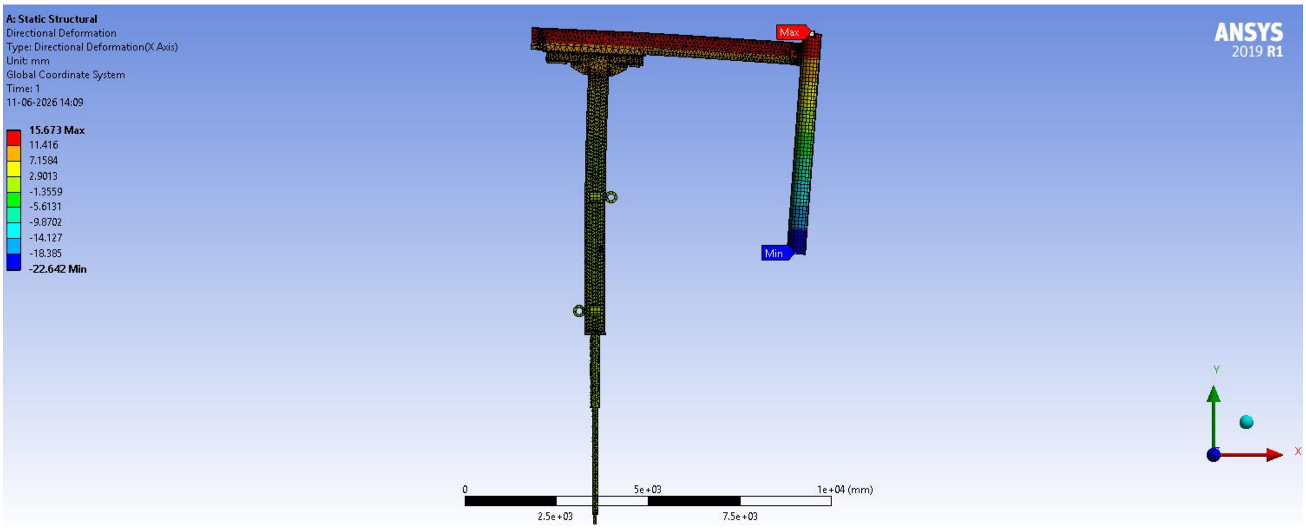
21. Equivalent Stress



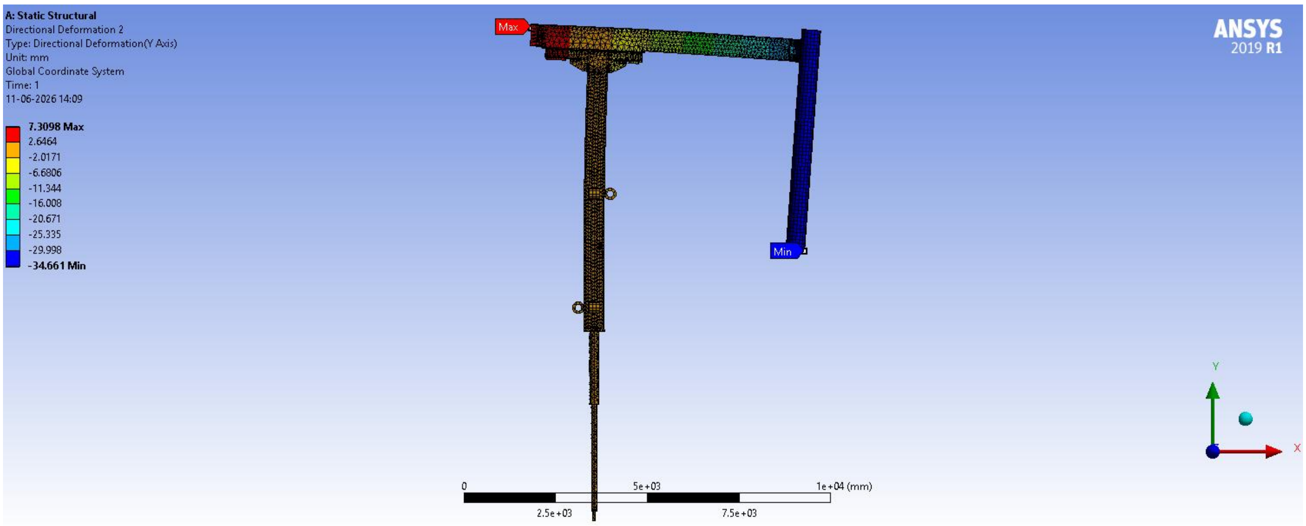
22. Total Deformation



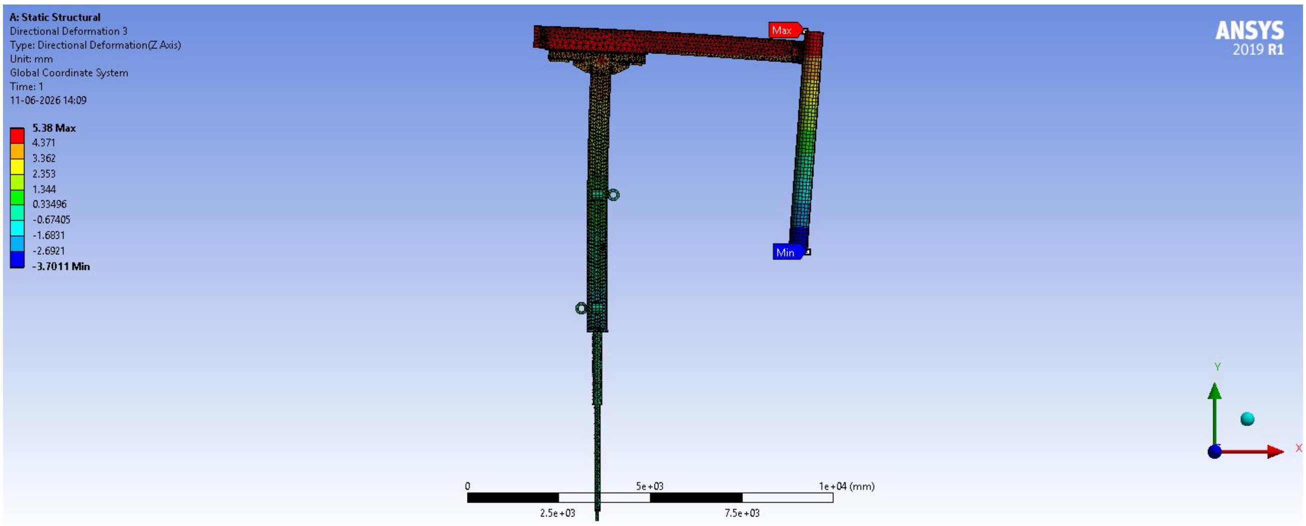
23. Deformation probes



24. X-axis Deformation



25. Y-axis Deformation



26. Z-axis Deformation

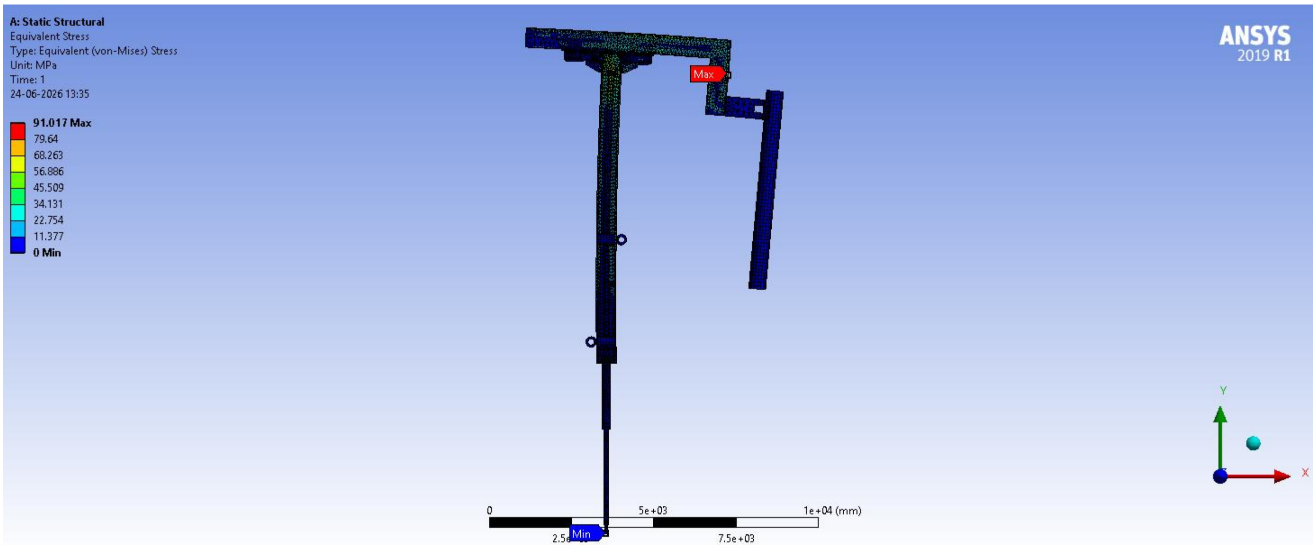
9.2 E-ARM#2 (Middle Arm) – Ø508 mm Electrode

Note: weight of the online electrode jointer is not considered. Roller vertical spacing: 3100 mm.

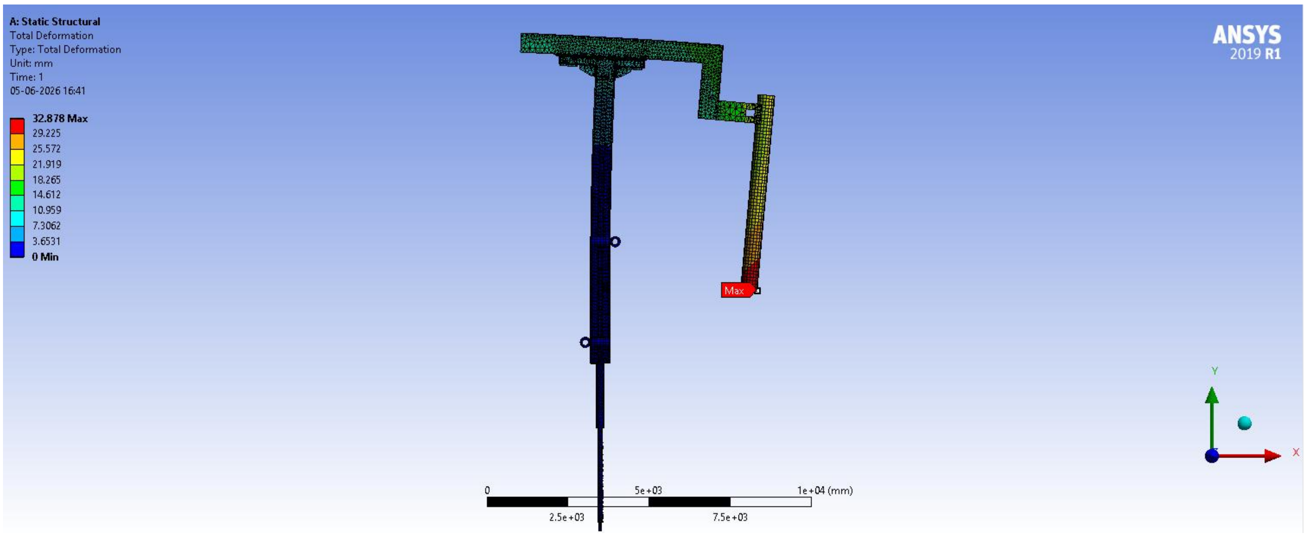
9.2.1 Mast 1 (Original: 600 mm × 410 mm)

Loads: Upper chamber water 8927.5 N, Lower chamber water 8723 N, Mast water (W3, W4, W5) 614.07 N / 698.62 N / 3521 N, Dust load (no FOS), Water-cooled cables + water 19958 N, Slag on electrode arm (no FOS), Electrode de-clamping unit weight 9335.2 N, Pressure 0.9 MPa (9 bar) applied on the side surfaces of the lower and upper chambers.

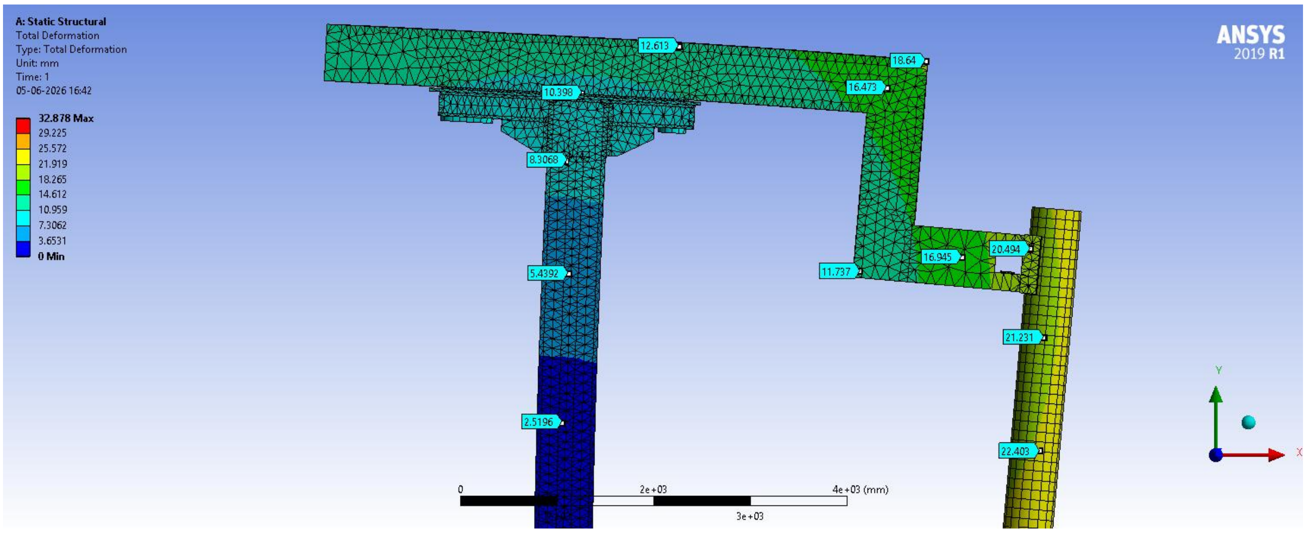
Equivalent (Von Mises) Stress – Max	91.017 MPa
Total Deformation – Max (at electrode tip)	32.878 mm
Deformation – Electrode Arm (range)	12 – 18 mm (max 20.5 mm at gripping point)
Directional Deformation X – Max	23.897 mm
Directional Deformation Y – Max	22.587 mm
Reaction Force – Upper Roller (–ve x-direction)	95,966 N
Reaction Force – Lower Roller (+ve x-direction)	95,821 N



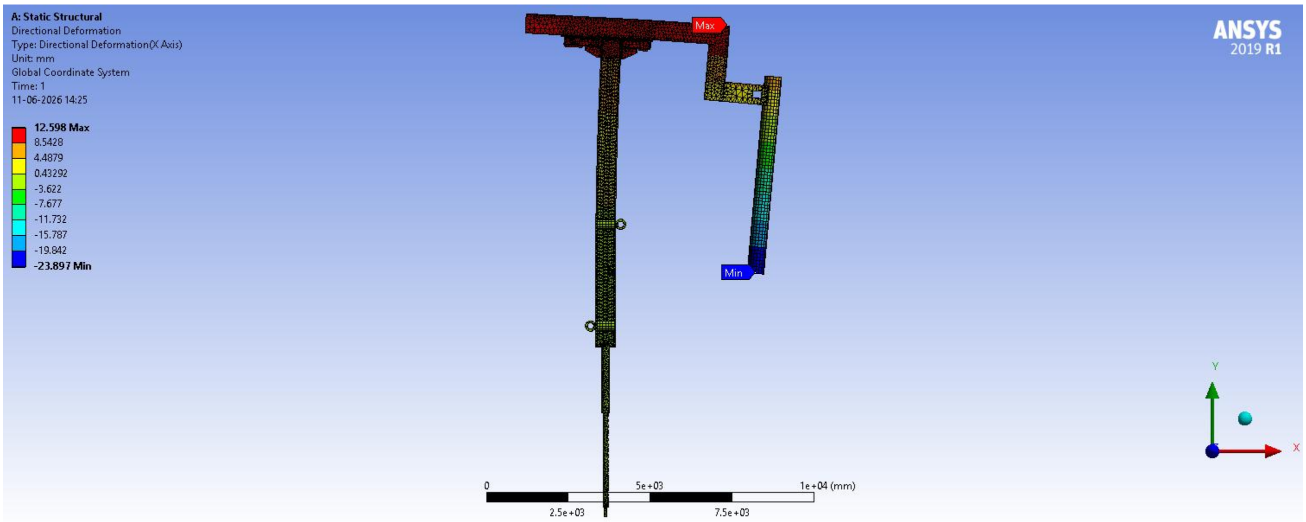
27. Equivalent Stress



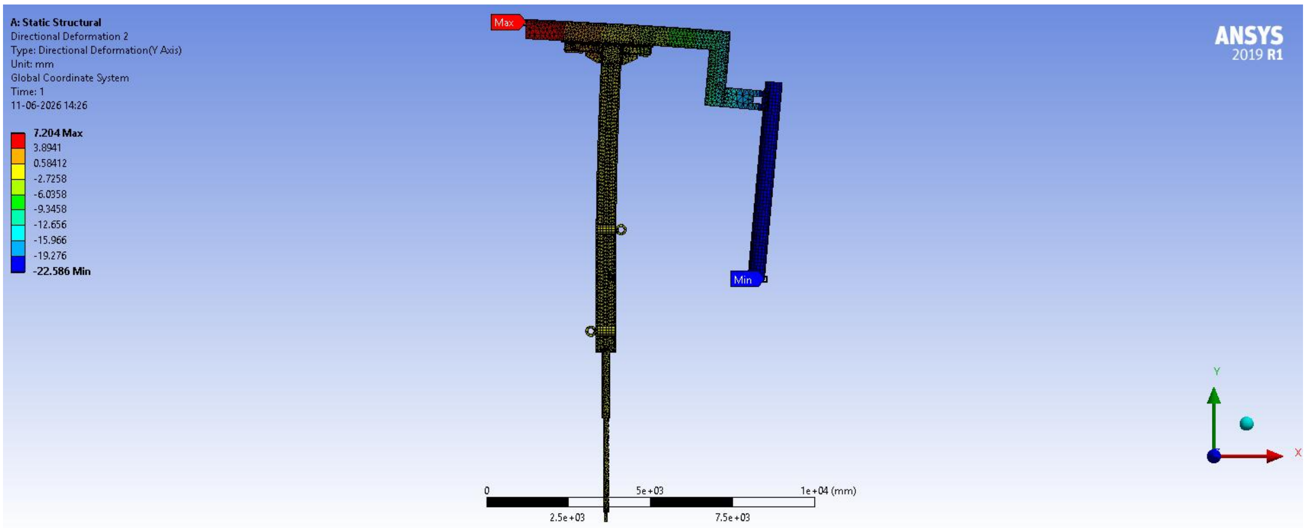
28. Total Deformation



29. Deformation Probes



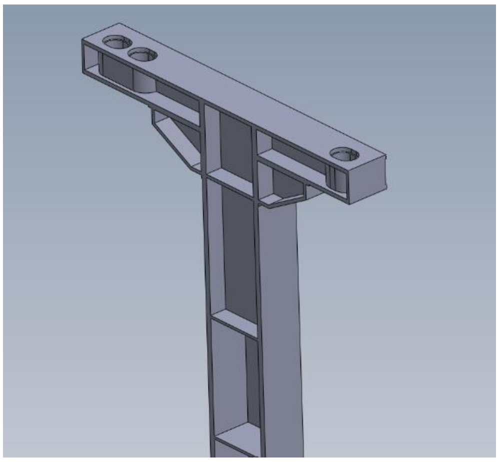
30. X-axis Deformation



31. Y-axis Deformation

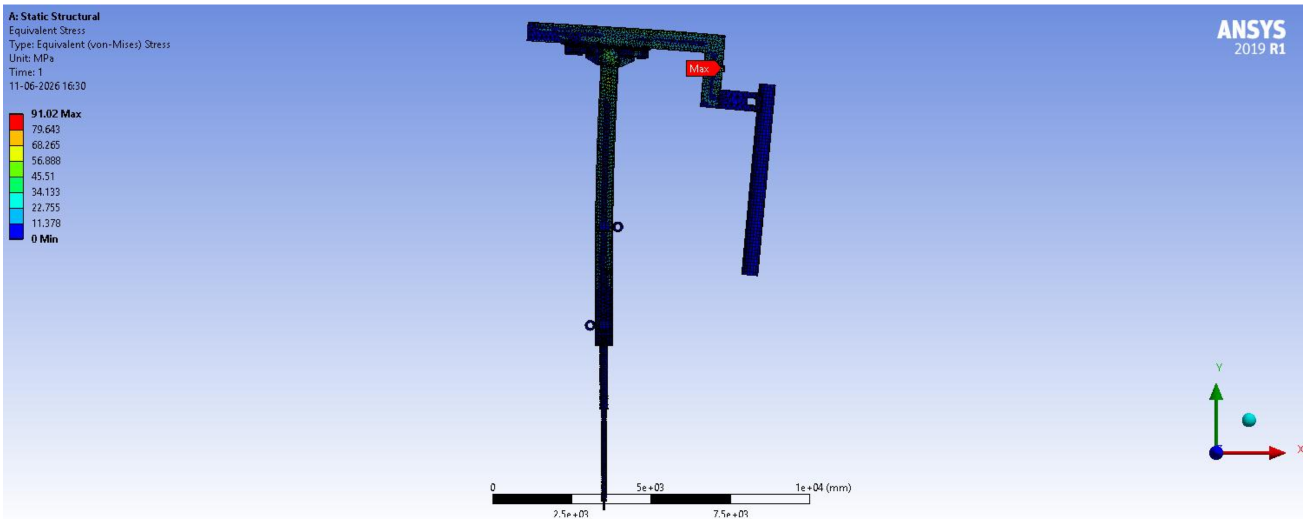
9.2.2 Mast 2 (Redesigned: 550 mm × 350 mm, plate added at mast top)

Cross-section altered from 600 × 410 mm to 550 × 350 mm; a plate was added to the top of the mast to counter stress. All electrode-arm load conditions unchanged. Mast water loads updated: W3 (614.07 N → 527.11 N), W4 (698.62 N → 581.47 N), W5 (3521 N → 1768.1 N); the added top plate exerts an additional 905.82 N due to the water column.

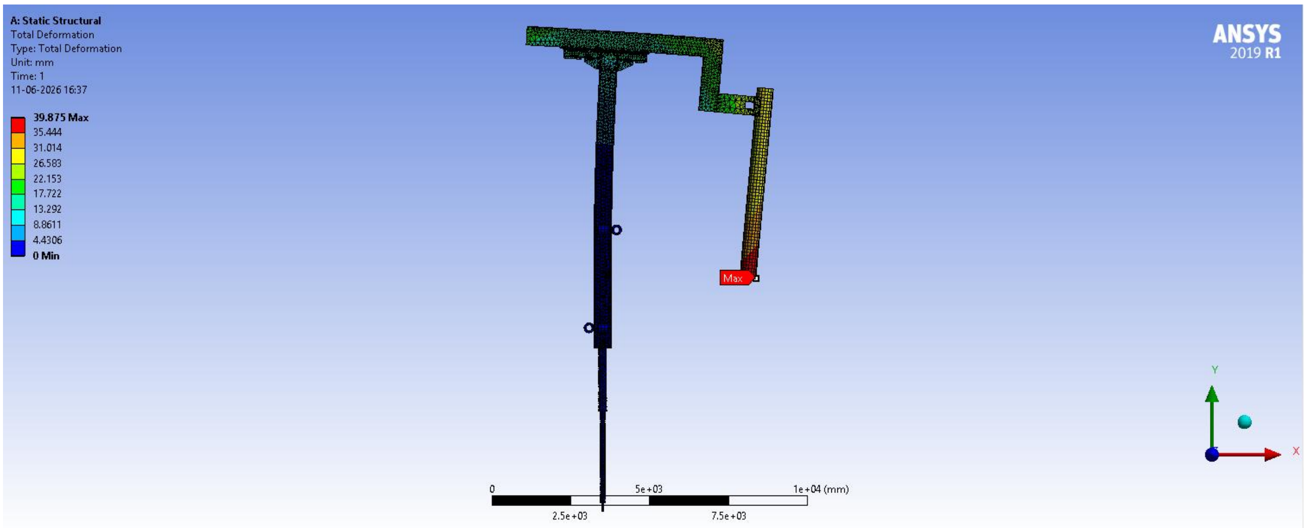


32. Added an extra plate in the mast

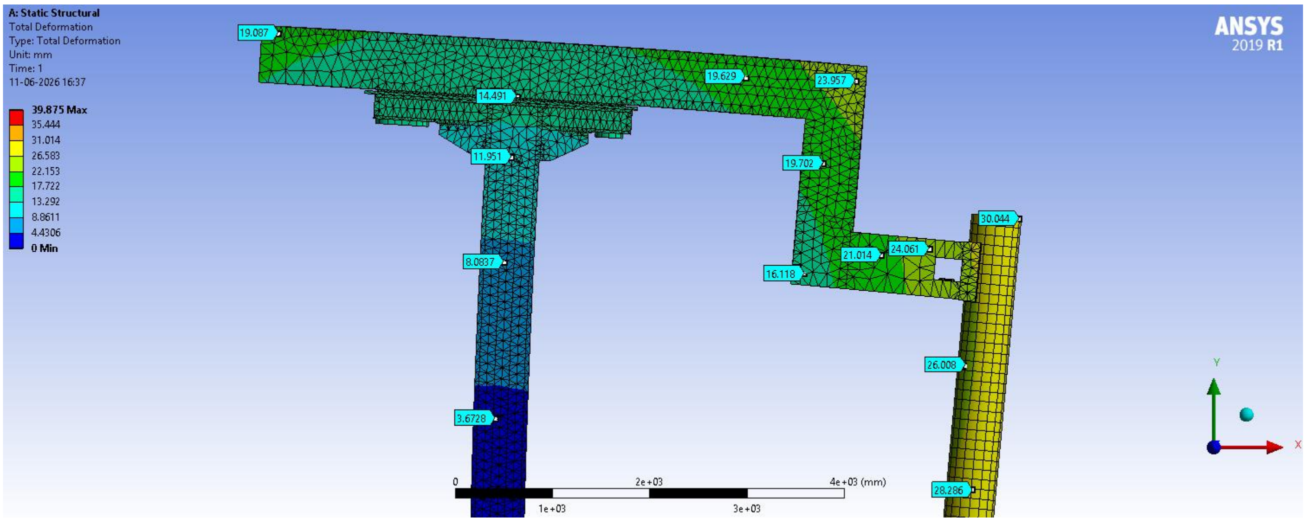
Equivalent (Von Mises) Stress – Max	91.02 MPa
Total Deformation – Max (at electrode tip)	39.875 mm
Deformation – Electrode Arm (range)	18 – 24 mm (max 30 mm at gripping point)
Directional Deformation X – Max	27.71 mm
Directional Deformation Y – Max	28.674 mm
Reaction Force – Upper Roller (–ve x-direction)	95,898 N
Reaction Force – Lower Roller (+ve x-direction)	95,723 N



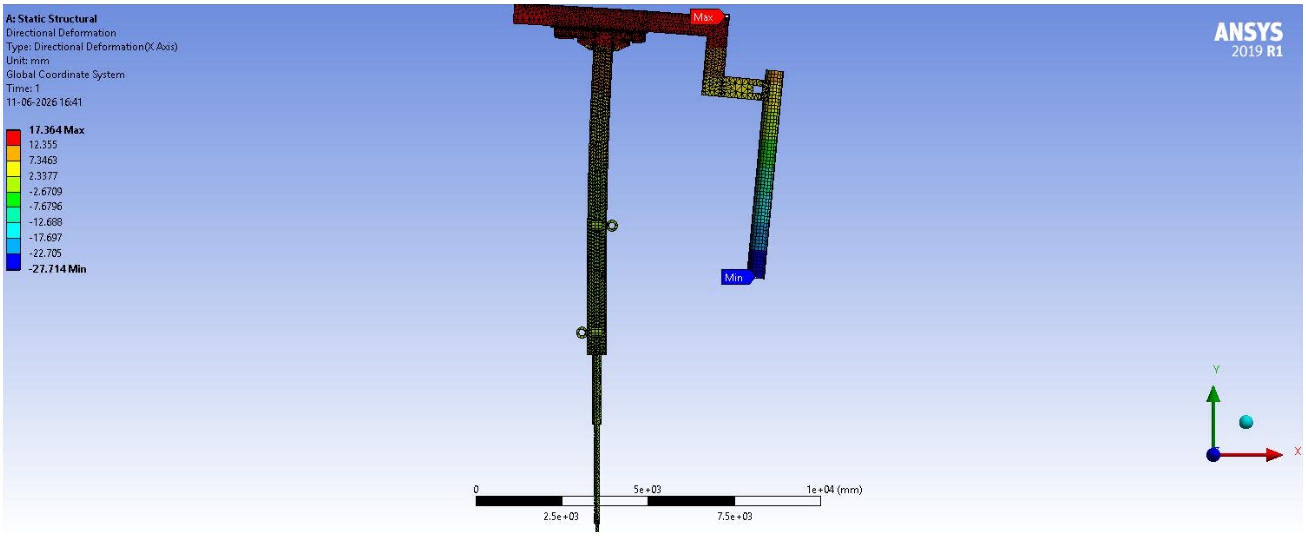
33. Equivalent Stress



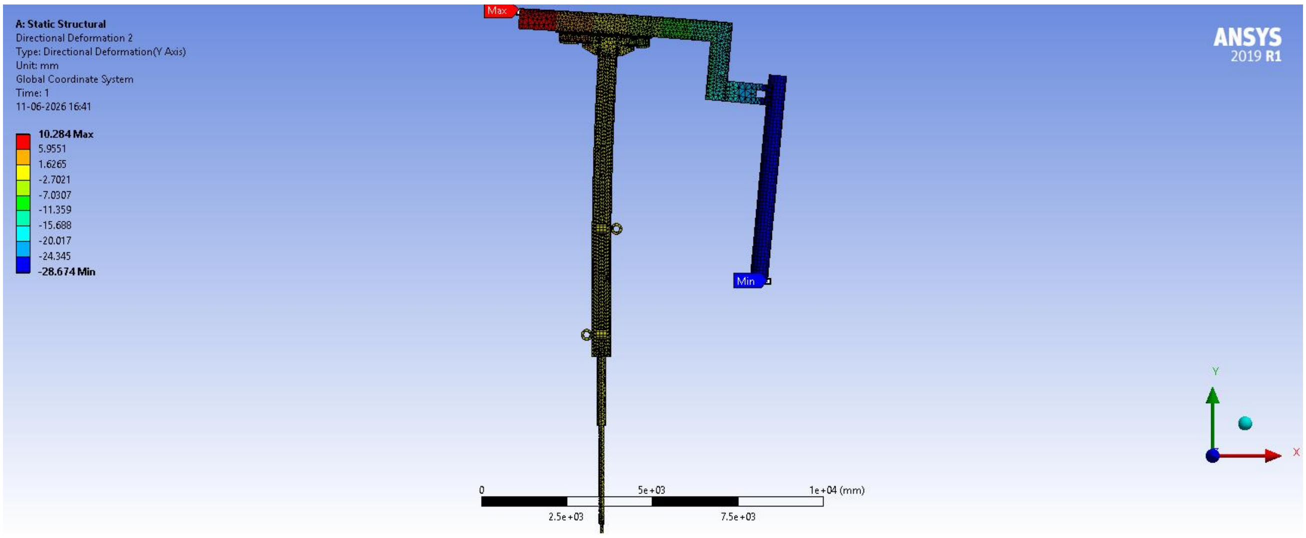
34. Total Deformation



35. Deformation Probes



36. X-axis Deformation



37. Y-axis Deformation

9.3 E-ARM#1 (Side Arm) – Ø457 mm Electrode

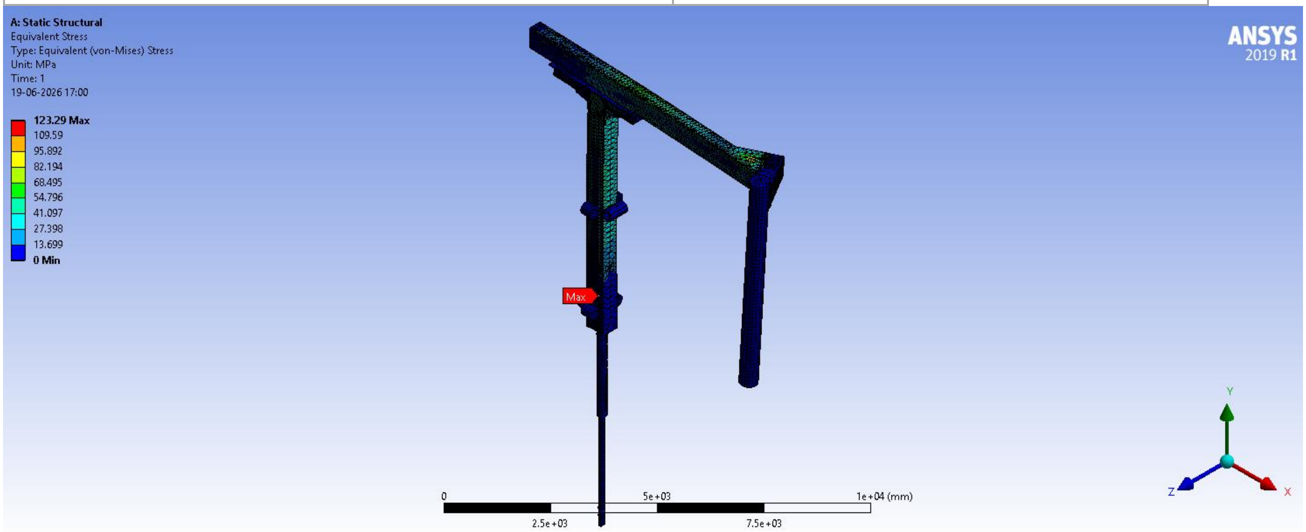
Roller vertical spacing: 2900 mm.

9.3.1 Mast 1 (Original: 550 mm × 400 mm)

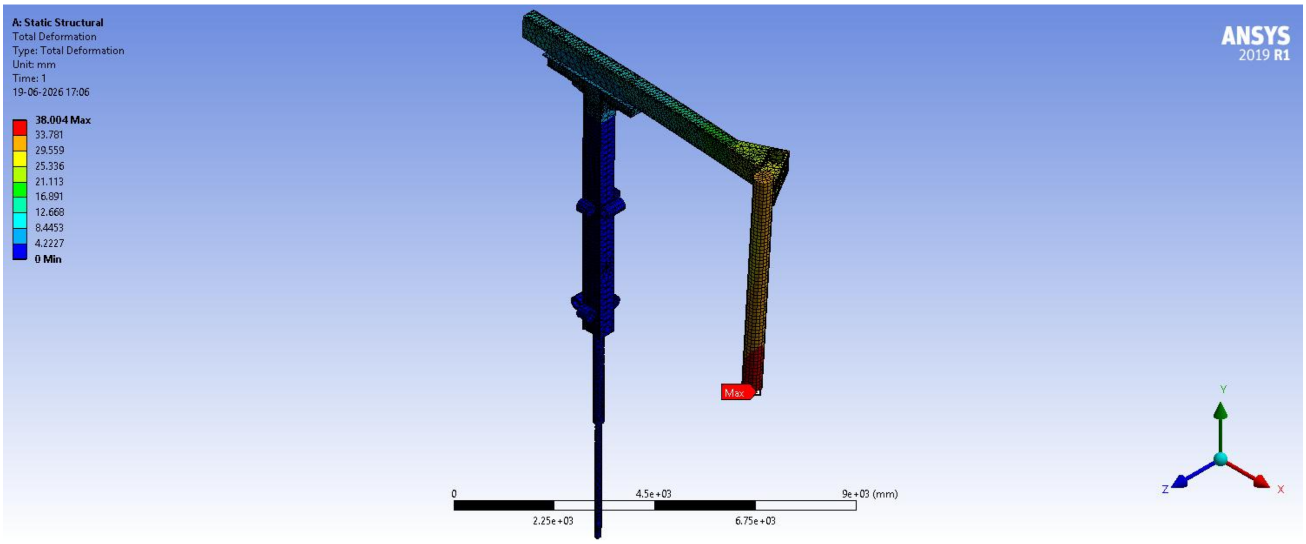
Loads: Upper chamber water 7714.6 N, Lower chamber water 7299.1 N, Mast water 611.69 N / 690.2 N / 3113.7 N, Dust load (no FOS), Cables 19961 N, Slag (no FOS), Clamping unit 9335.2 N, Pressure 0.9 MPa (9 bar) on the side surfaces of the lower and upper chambers. Note: excludes weight of the online electrode jointer.

Equivalent (Von Mises) Stress – Max	123.29 MPa
Total Deformation – Max	38.004 mm
Directional Deformation X – Max	23.302 mm
Directional Deformation Y – Max	29.786 mm
Directional Deformation Z – Max	3.765 mm

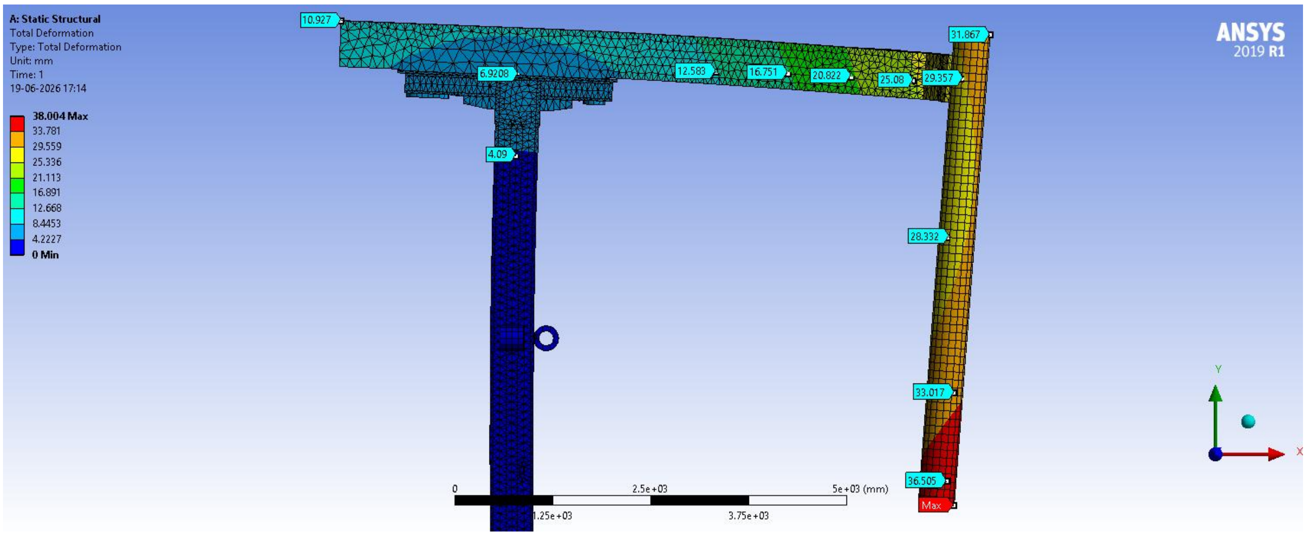
Reaction Force – Upper Roller	135,330 N
Reaction Force – Lower Roller	136,130 N
Total Load – Hydraulic Cylinder Base	175,360 N



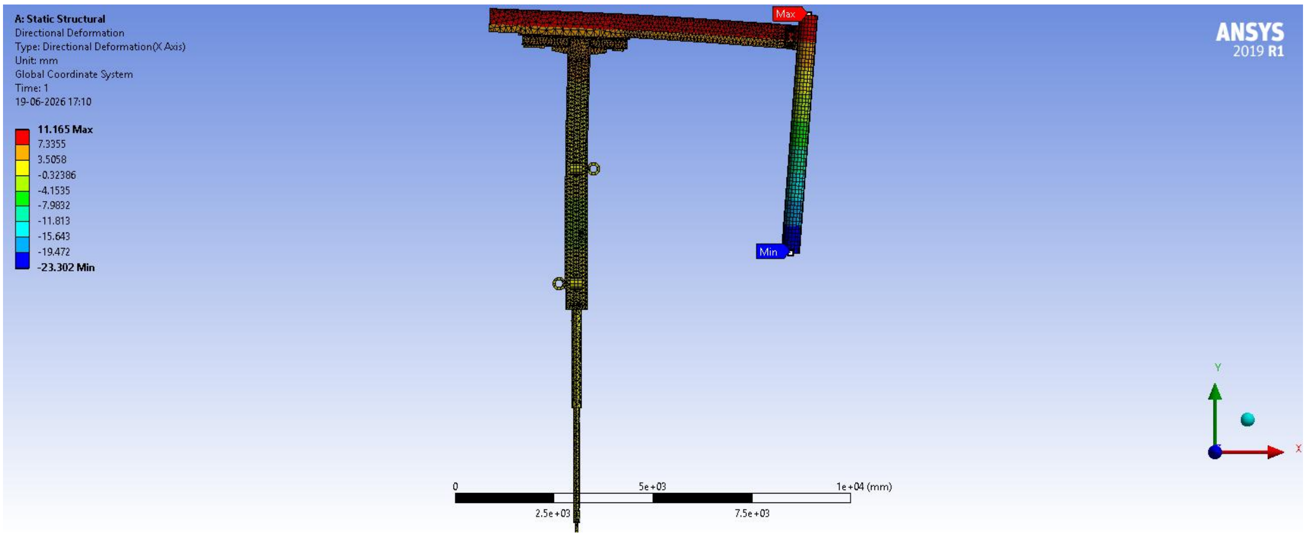
38. Equivalent Stress



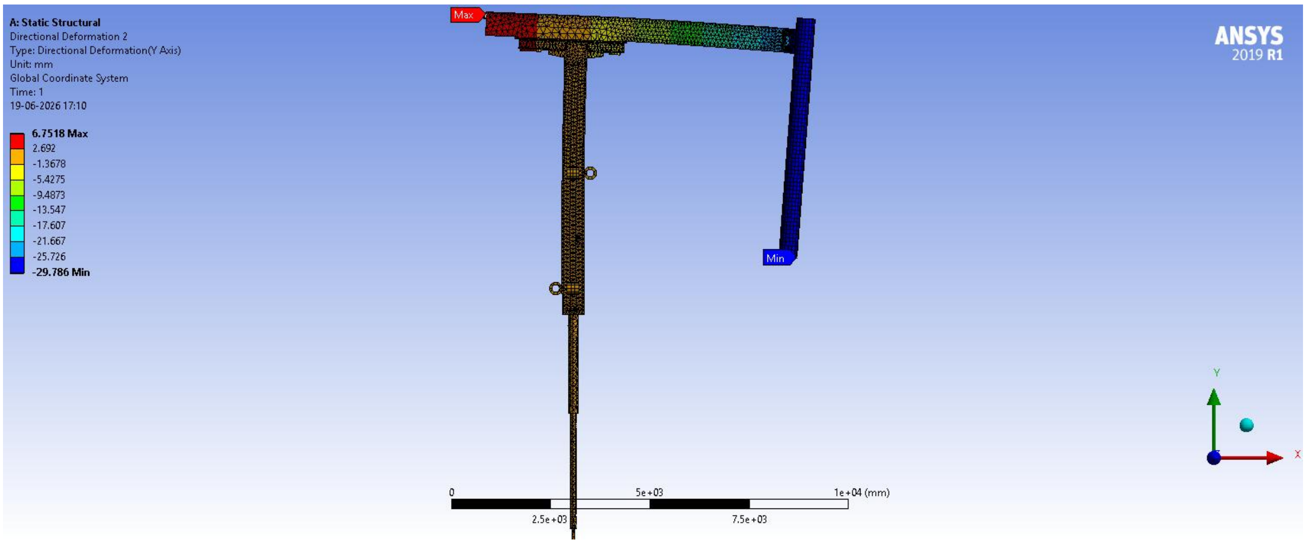
39. Total Deformation



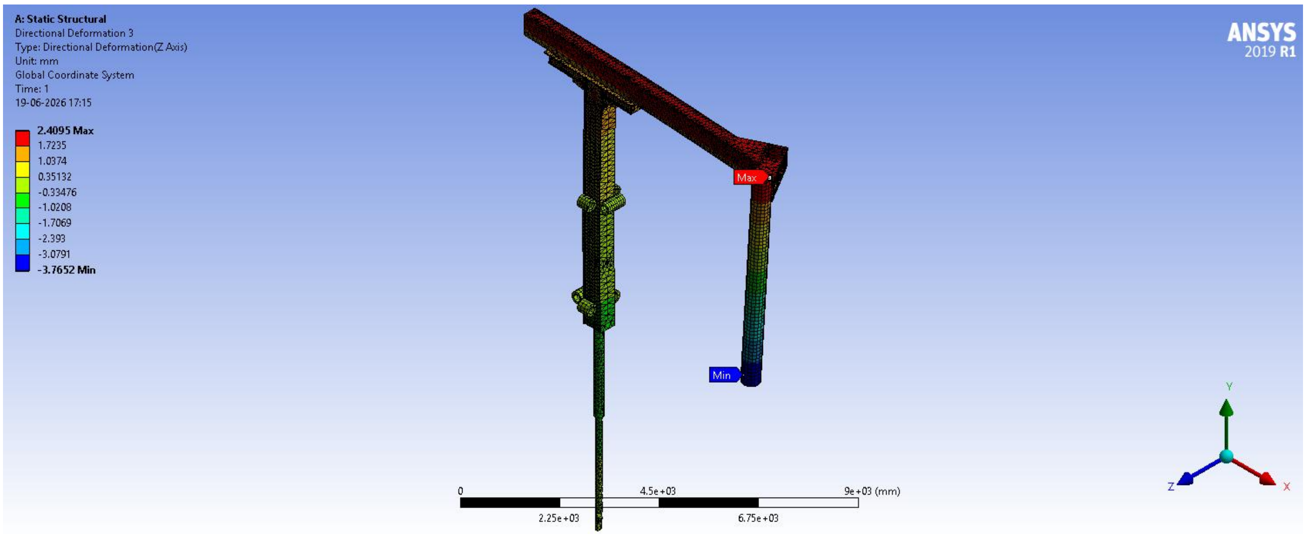
40. Deformation Probes



41. X-axis Deformation



42. Y-axis Deformation



43. Z-axis Deformation

9.3.2 Mast 2 (Redesigned – proposed)

By the time the Ø457 mm side arm (E-ARM#1) was analyzed, optimized designs had already been validated for both the Ø508 mm side arm and the Ø457 mm middle arm (E-ARM#2). Based on this experience, it was concluded that the same redesign approach validated for the Ø457 mm middle arm — cross-section reduced from 550 × 400 mm to 500 × 310 mm, with the major-axis plate thickness increased from 40 mm to 42 mm and the minor-axis plate thickness reduced from 25 mm to 20 mm — could be directly applied to the Ø457 mm side arm to achieve similarly favourable results (reduced stress and base load with an acceptable increase in deformation), without requiring a fresh redesign-and-optimization cycle.

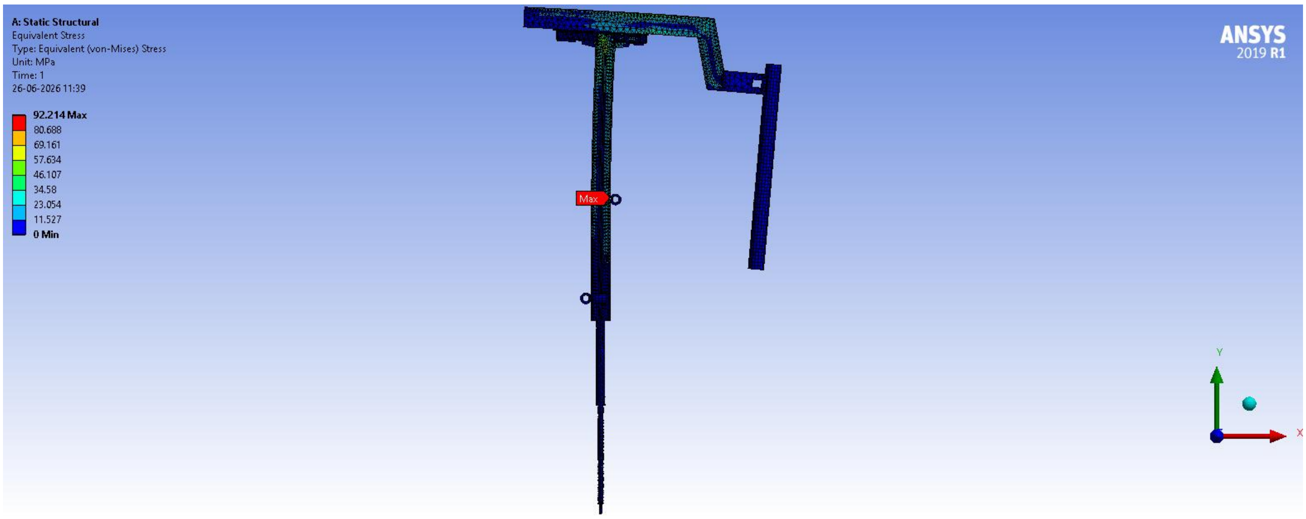
9.4 E-ARM#2 (Middle Arm) – Ø457 mm Electrode

Roller vertical spacing: 2900 mm.

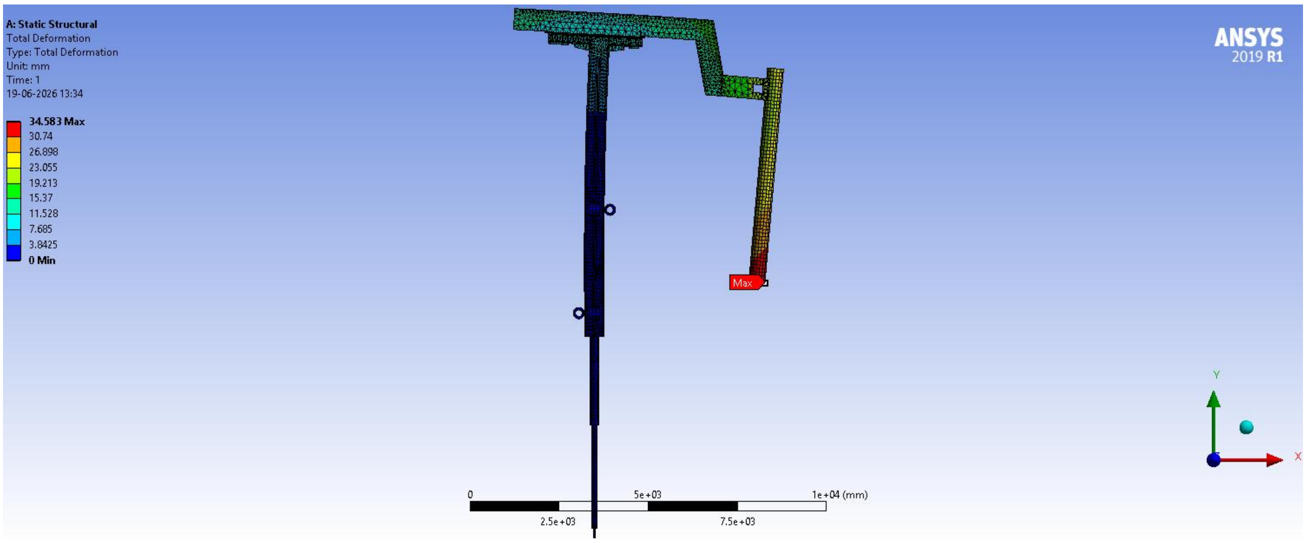
9.4.1 Mast 1 (Original: 550 mm × 400 mm)

Loads: Upper chamber water 6873.3 N, Lower chamber water 6509.6 N, Mast water 611.69 N / 690.2 N / 3113.7 N, Dust load (no FOS), Cables 19961 N, Slag (no FOS), Clamping unit 9335.2 N, Pressure 0.9 MPa (9 bar) on the side surfaces of the lower and upper chambers.

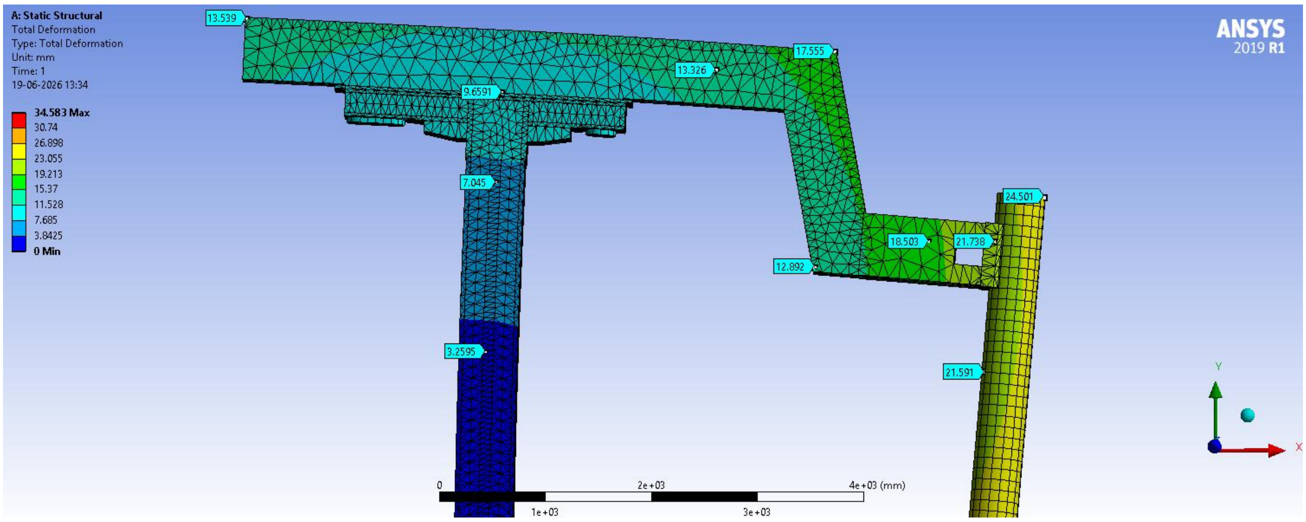
Equivalent (Von Mises) Stress – Max	92.214 MPa
Total Deformation – Max	34.583 mm
Directional Deformation X – Max	25.152 mm
Directional Deformation Y – Max	23.736 mm
Reaction Force – Upper Roller	102,020 N
Reaction Force – Lower Roller	101,980 N
Total Load – Hydraulic Cylinder Base	183,380 N



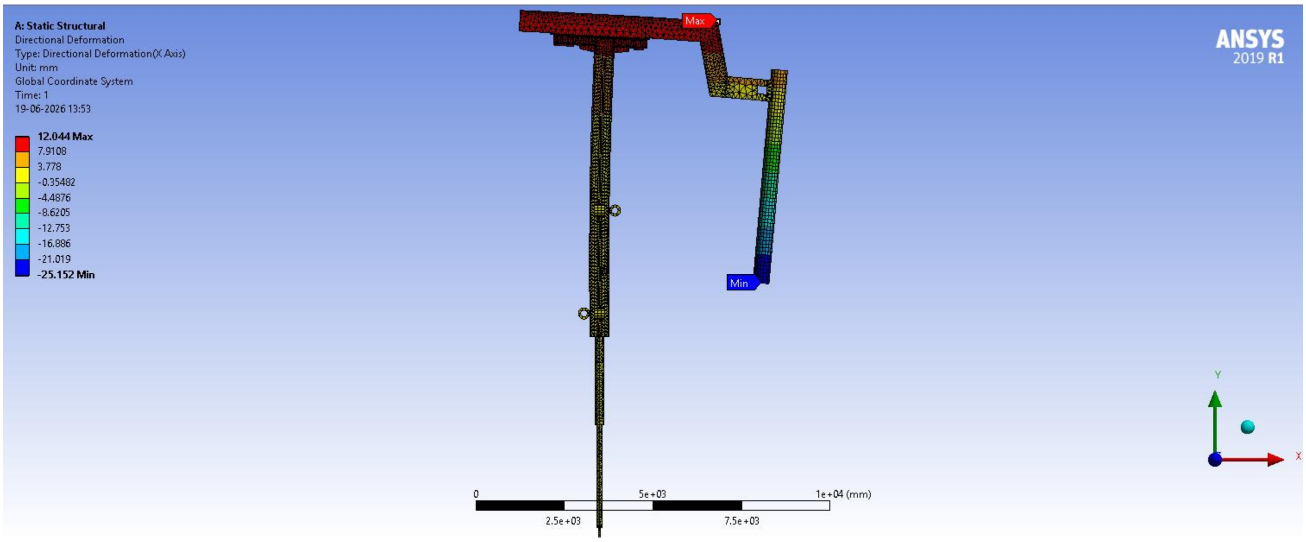
44. Equivalent Stress



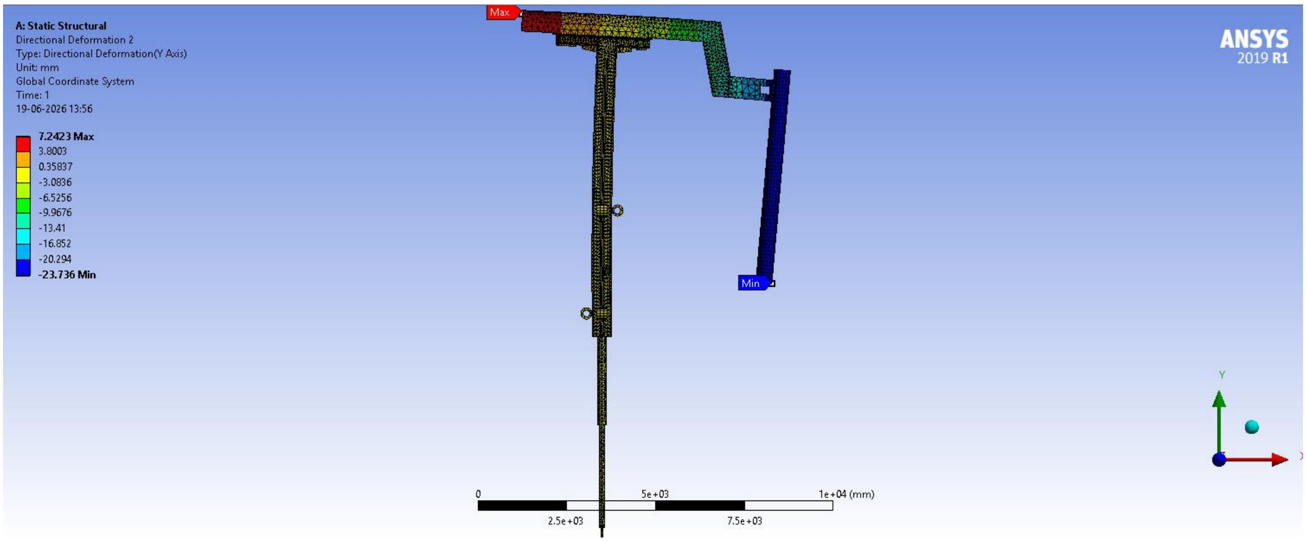
45. Total Deformation



46. Deformation Probes



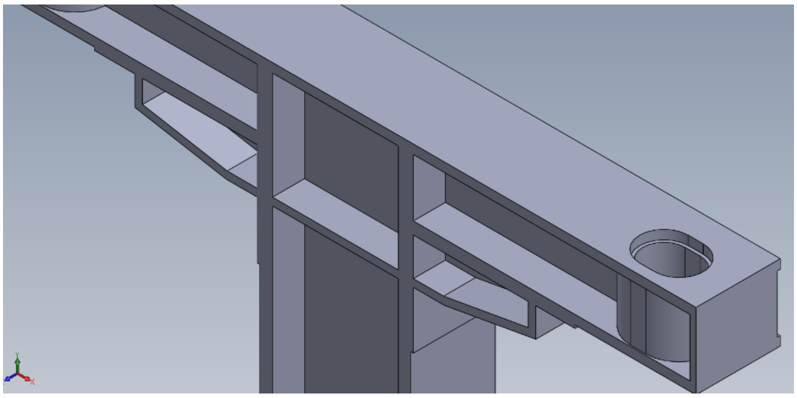
47. X-axis Deformation



48. Y-axis Deformation

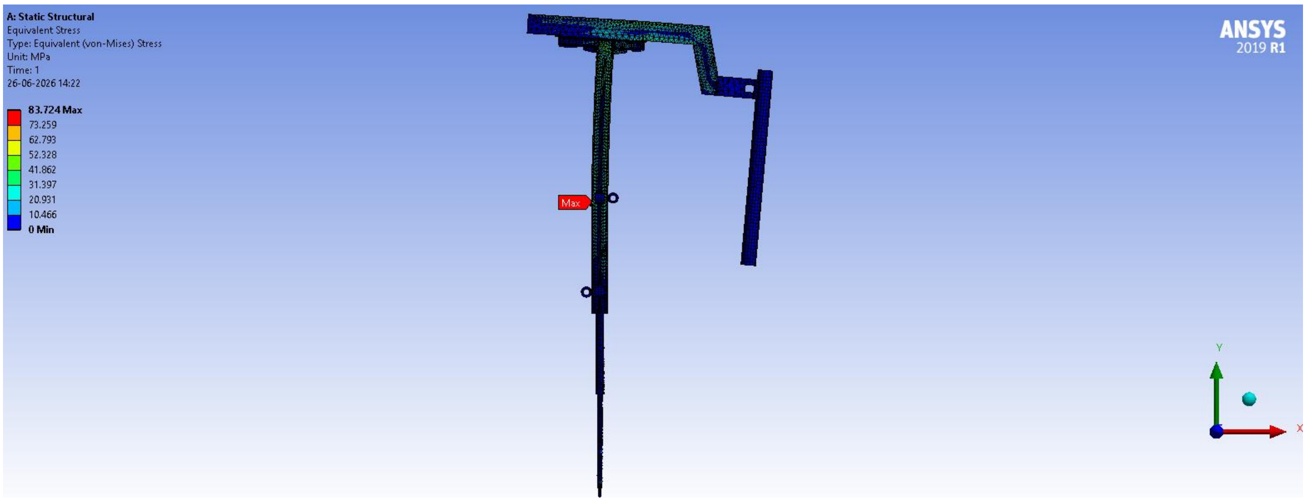
9.4.2 Mast 2 (Redesigned: 500 mm × 310 mm)

Cross-section reduced from 550 × 400 mm to 500 × 310 mm. The 40 mm major-axis plate thickness was increased to 42 mm; the 25 mm minor-axis plate thickness was reduced to 20 mm. Mast water weight recalculated for the new internal volumes: 6873.3 N, 6509.6 N, 487.68 N, 455.03 N, 1583.3 N, 476.94 N. All other forces (dust, cables, slag, clamping unit, pressure) unchanged from Mast 1.

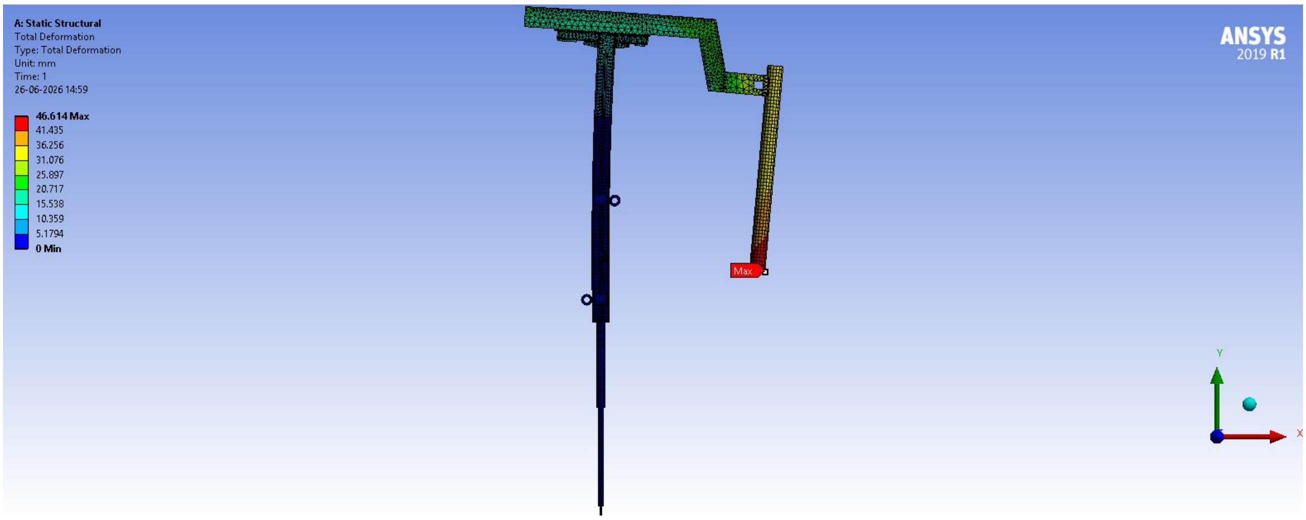


49. Added an extra plate in the mast

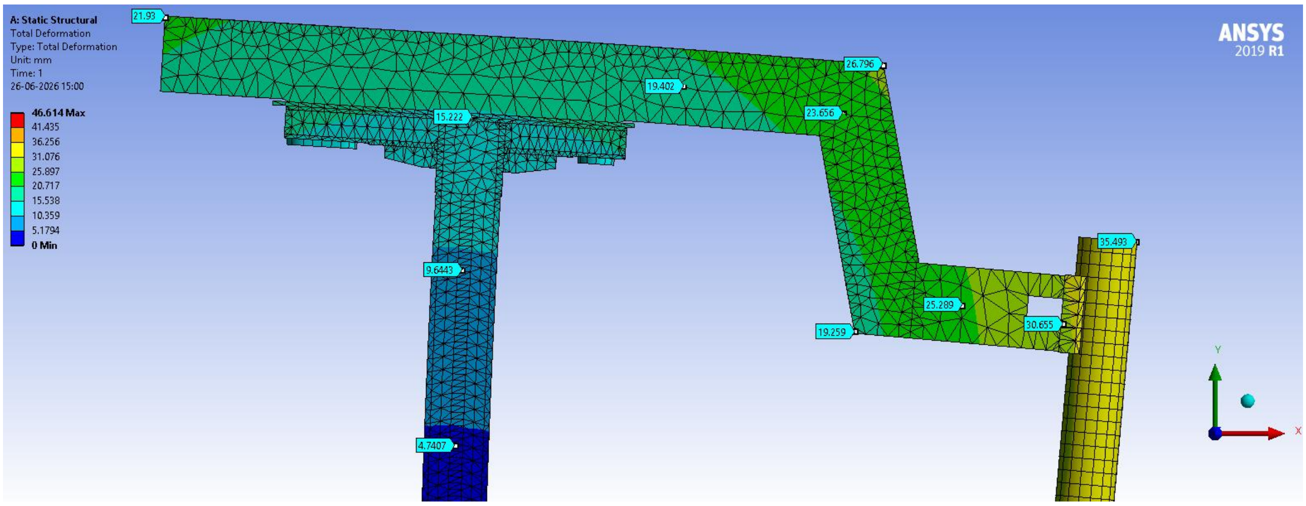
Equivalent (Von Mises) Stress – Max	83.724 MPa
Total Deformation – Max	46.614 mm
Directional Deformation X – Max	32.128 mm
Directional Deformation Y – Max	33.775 mm
Reaction Force – Upper Roller	103,690 N
Reaction Force – Lower Roller	104,700 N
Total Load – Hydraulic Cylinder Base	165,870 N (~9.5% reduction)



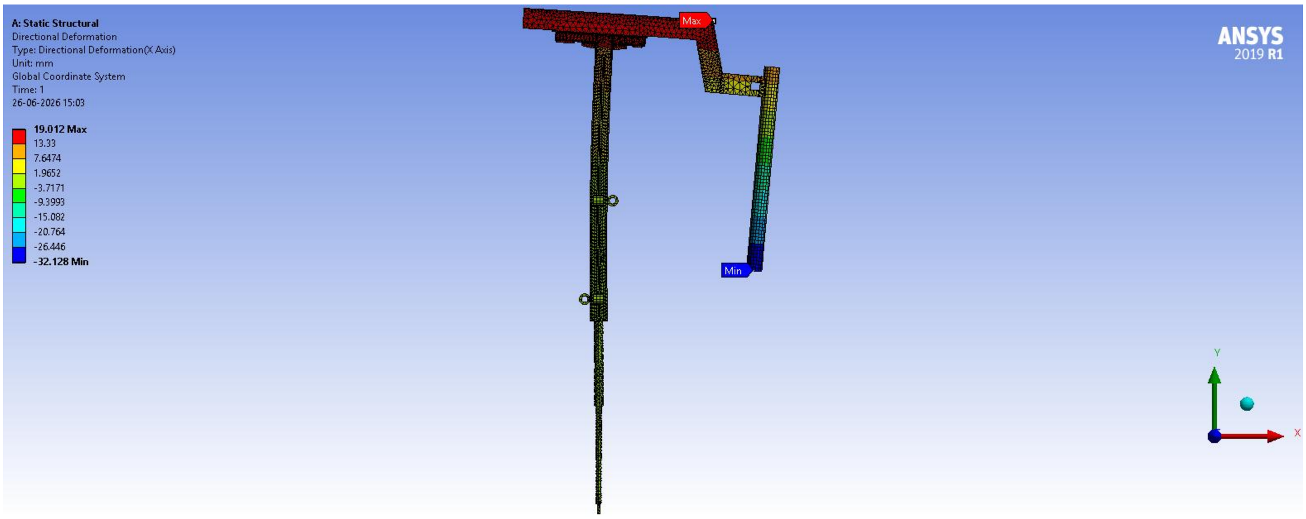
50. *Equivalent Stress*



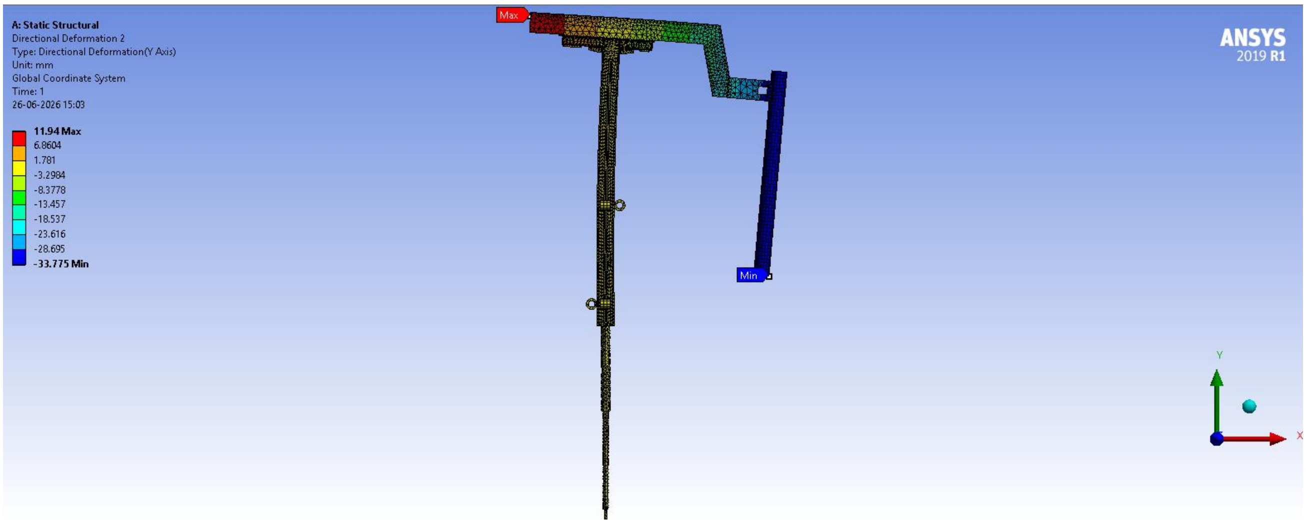
51. *Total Deformation*



52. *Deformation Probes*



53. X-axis Deformation

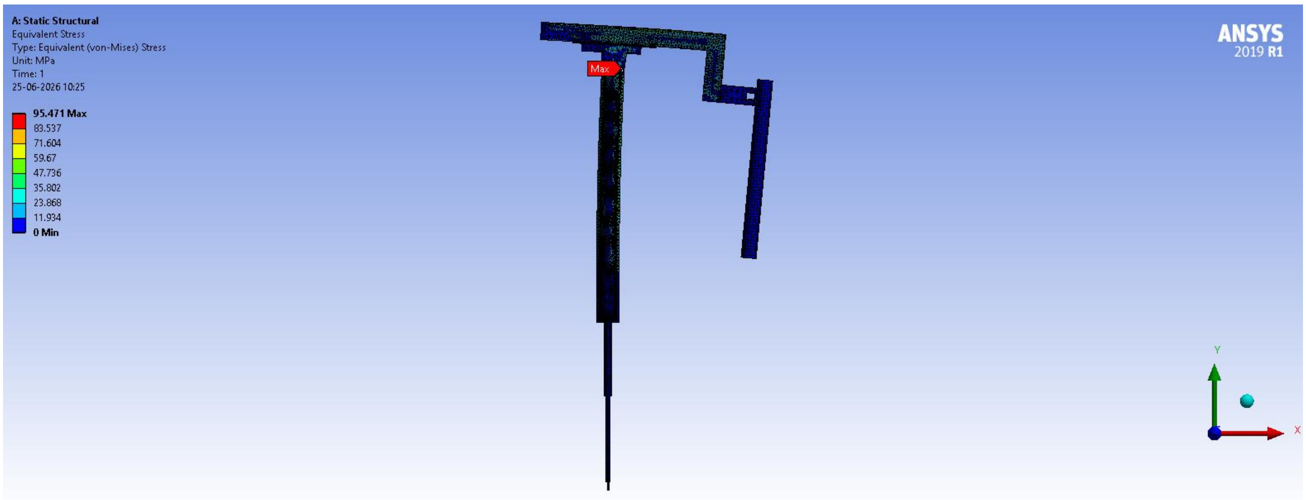


54. Y-axis Deformation

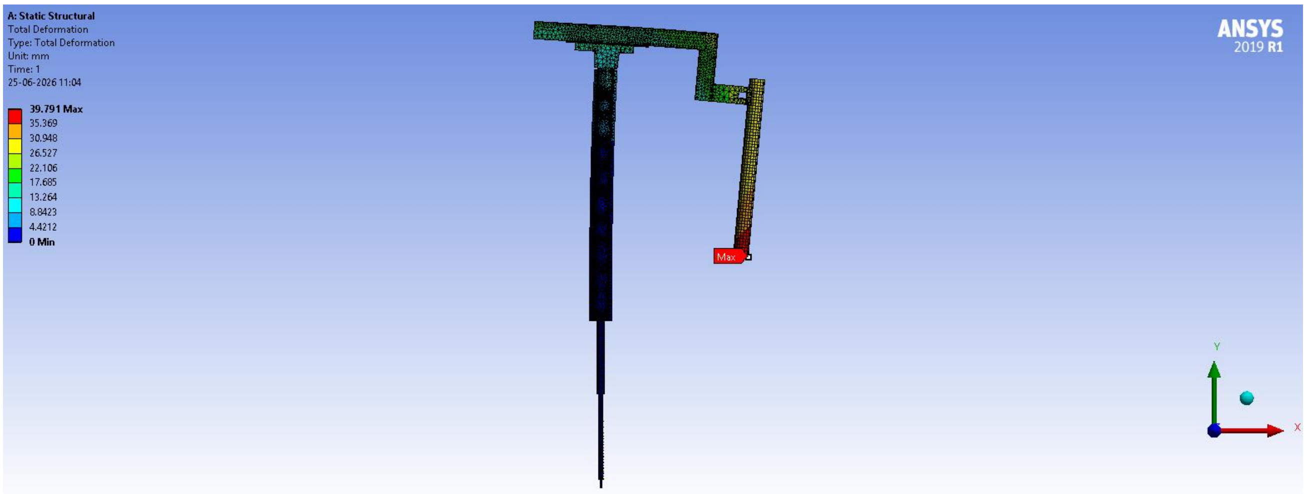
9.5 Cylindrical Mast Study

A circular hollow section (cylindrical) mast was modeled and analyzed as an alternative to the standard rectangular hollow section, using the same loading conditions, pressures, and fixed supports as the E-ARM#2 (Middle Arm), Ø508 mm electrode, Mast 1 configuration (Section 9.2.1). The mast water weights used were 8927.5 N, 8723 N, 583.63 N, 669.19 N, 4051.5 N, and 584.64 N.

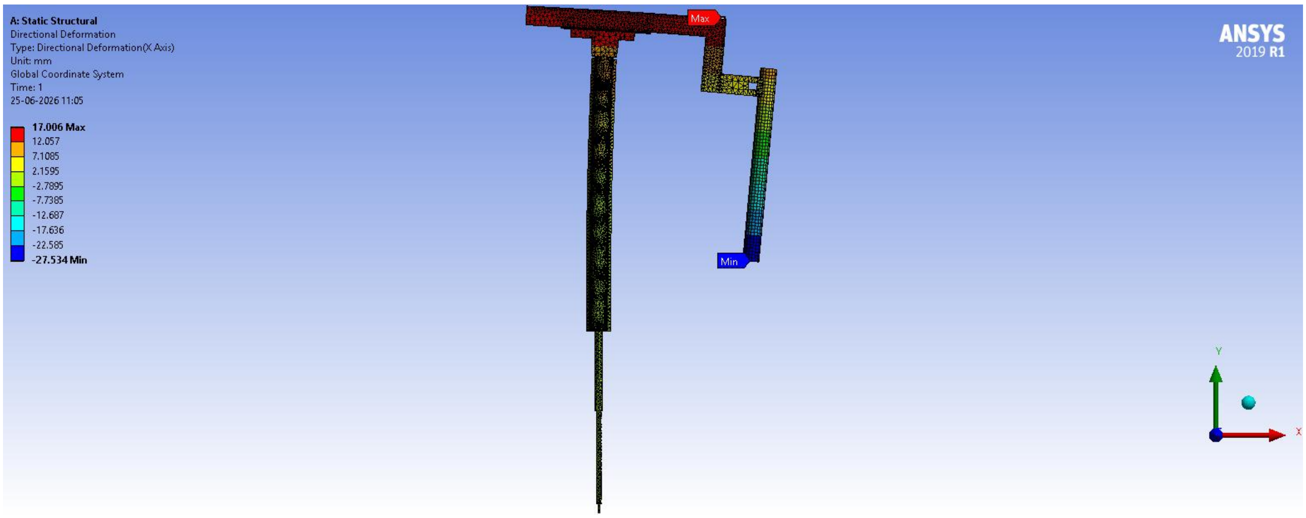
Equivalent (Von Mises) Stress – Max	95.471 MPa
Total Deformation – Max	39.791 mm
Directional Deformation X – Max	27.534 mm
Directional Deformation Y – Max	28.693 mm



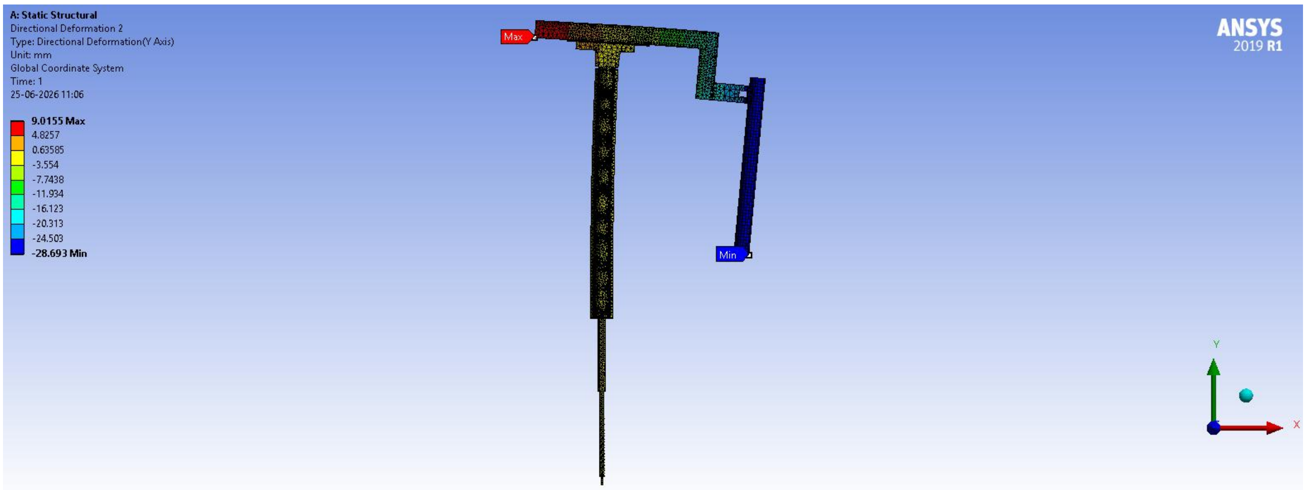
55. Equivalent Stress



56. Total Deformation



57. X-axis Deformation



58. Y-axis Deformation

- Compared to the rectangular Mast 1 for the same configuration (91.017 MPa, 32.878 mm), the cylindrical mast shows both higher peak stress (95.471 MPa) and higher total deformation (39.791 mm).
- This confirms a lower effective bending stiffness for the cylindrical cross-section compared to the equivalent rectangular section, for the same loading conditions.
- Conclusion: The cylindrical mast concept was found structurally inferior to the rectangular section and was not pursued further.

10. Comparison and Weight Reduction Summary

10.1 E-ARM#1 (Side Arm) – Ø508 mm Electrode

Parameter	Mast 1 (600 × 410 mm)	Mast 2 (550 × 350 mm)
Von Mises Stress – Max (in mast)	131.5 MPa	102.88 MPa (−21.76%)
Total Deformation – Max	34.247 mm	41.56 mm (+21.4%)
Hydraulic Cylinder Base Load	192.48 kN	178.84 kN (−7.1%)
Verdict	Baseline	Improved – stress and base load both reduced

For the side arm (E-ARM#1, Ø508 mm electrode), the redesigned Mast 2 reduces both peak stress (~21.76%) and base load (~7.1%), confirming a genuine structural improvement despite the moderate increase in tip deformation.

10.2 Weight and Cost Reduction Summary

The mast weights for each configuration, obtained from SolidWorks mass properties, are summarized below along with the resulting weight and cost savings. The material cost is estimated at ₹400/kg — this reflects the cost of processed and fabricated structural steel (S355), accounting for cutting, welding, forming, and surface treatment, and is not simply the raw material price.

Configuration	Mast 1 Weight (kg)	Mast 2 Weight (kg)	Weight Reduction	Cost Saving
E-ARM#1, Ø508 mm	5309	4518.9	790.1 kg (14.9%)	≈ ₹3.16 lakhs
E-ARM#2, Ø508 mm	6234	5274.8	959.2 kg (15.4%)	≈ ₹3.84 lakhs
E-ARM#1, Ø457 mm	4663.3	3609.8 (proposed)	1053.5 kg (22.6%)	≈ ₹4.21 lakhs
E-ARM#2, Ø457 mm	6221.4	4951.17	1270.2 kg (20.4%)	≈ ₹5.08 lakhs

Across all four configurations, the weight reduction ranges from approximately 15% to 23%, with the corresponding cost saving ranging from roughly ₹3.2 lakhs to ₹5.1 lakhs per mast (at ₹400/kg for processed fabricated steel). These figures represent the core quantified outcome of the project. Note that for E-ARM#1 (Ø457 mm), the Mast 2 figures are based on the proposed redesign described in Section 9.3.2, applied by analogy with the validated E-ARM#2 (Ø457 mm) results, rather than a dedicated ANSYS re-analysis.

10.3 E-ARM#2 (Middle Arm) – Ø508 mm Electrode

Parameter	Mast 1 (600 × 410 mm)	Mast 2 (550 × 350 mm)
Von Mises Stress – Max	91.017 MPa	91.02 MPa
Total Deformation – Max	32.878 mm	39.875 mm (+21%)

Mast Weight	6234 kg	5274.8 kg (−15.4%)
Verdict	Baseline	Peak stress essentially unchanged; mast weight reduced by ~15.4%

The redesigned middle-arm mast achieves an almost identical peak stress (~91 MPa for both) while reducing mast weight by ~15.4%, confirming that the added top plate effectively compensates for the reduced cross-section.

10.4 E-ARM#1 (Side Arm) – Ø457 mm Electrode

Parameter	Mast 1 (550 × 400 mm)
Von Mises Stress – Max	123.29 MPa
Total Deformation – Max	38.004 mm
Mast Weight	4663.3 kg
Hydraulic Cylinder Base Load	175,360 N

As noted in Section 9.3.2, a dedicated redesign-and-reanalysis cycle was not performed for this configuration. Instead, based on the favourable results already validated for the Ø508 mm side arm and the Ø457 mm middle arm, it was concluded that the same redesign (550 × 400 mm → 500 × 310 mm, with the 40 mm plate increased to 42 mm and the 25 mm plate reduced to 20 mm) could be applied directly to the Ø457 mm side arm to achieve comparable stress and weight reductions. The estimated Mast 2 weight, based on this proposed redesign, is 3609.8 kg — a reduction of ≈1053.5 kg (22.6%) and an estimated cost saving of ≈₹4.21 lakhs.

10.5 E-ARM#2 (Middle Arm) – Ø457 mm Electrode

Parameter	Mast 1 (550 × 400 mm)	Mast 2 (500 × 310 mm)
Von Mises Stress – Max	92.214 MPa	83.724 MPa (−9.2%)
Total Deformation – Max	34.583 mm	46.614 mm (+34.8%)
Mast Weight	6221.4 kg	4951.17 kg (−20.4%)
Hydraulic Cylinder Base Load	183,380 N	165,870 N (−9.5%)
Verdict	Baseline	Stress, load, and weight all reduced; deformation increase within allowable limits

The redesigned mast (500 × 310 mm with revised plate thicknesses: 42 mm major-axis, 20 mm minor-axis) achieves a meaningful reduction in peak stress (~9.2%), base load (~9.5%), and weight (~20.4%) — the largest weight saving among all configurations studied. The corresponding increase in total deformation (~35%) remains within the project's allowable deflection limit.

10.6 Overall Summary – All Four Configurations

Configuration	Mast 1 X-sec	Mast 2 X-sec	Stress Change	Weight Change	Base Load Change
E-ARM#1, Ø508 mm	600×410	550×350	−21.76%	−14.9%	−7.1%
E-ARM#2, Ø508 mm	600×410	550×350	≈ 0%	−15.4%	≈ 0% (roller reactions)
E-ARM#1, Ø457 mm	550×400	500×310 (proposed)	Expected ≈ −9%	−22.6%	Expected ≈ −9.5%
E-ARM#2, Ø457 mm	550×400	500×310	−9.2%	−20.4%	−9.5%

Across all configurations studied, the reduced mast cross-section achieves equal or lower peak stress while reducing mast weight by approximately 15–23% and the load on the hydraulic cylinder base by roughly 7–10%, confirming a genuine and consistent reduction in structural weight across the product range. Cost savings, estimated at ₹400/kg for processed fabricated steel, range from roughly ₹3.2 lakhs to ₹5.1 lakhs per mast. The trade-off in all redesigned cases is an increase in total deformation, which in every case remains within the allowable deflection limit for the assembly.

11. Conclusions

This 6-week internship project at SMS India Pvt. Ltd. demonstrated a structured FEA-based approach to evaluate and redesign the electrode mast of a Ladle Heating Furnace for weight and cost reduction. Key conclusions:

- The complete electrode mast-arm assembly (mast, arm, electrode, insulating plate, rollers, hydraulic cylinder) was successfully modelled in SolidWorks with simplification, density adjustment, and FOS-based conservative loading.
- Bolt pretension has a negligible effect on global deformation and stress fields, allowing computationally efficient bonded-contact models for this scale of analysis.
- 100 mm mesh element size is adequate for global structural response of these assemblies (mast height 6.8–9.5 m, total assembly height up to ~15 m), confirmed by a mesh sensitivity study ($\Delta < 0.2$ mm at 75 mm mesh).
- The 8-roller guide system (4 pairs: front/back, left/right, upper/lower) is correctly modelled with the moment-resisting roller pair given fixed support and a frictional contact ($\mu = 0.03$) against the mast, while the opposite thermal-gap roller pair remains disengaged to accommodate thermal expansion. Roller vertical spacing is 3100 mm for the Ø508 mm electrode and 2900 mm for the Ø457 mm electrode.
- For E-ARM#1 (Ø508 mm electrode, side arm): The redesigned mast (550 × 350 mm, with a stiffener plate added at the mast top) reduces peak stress by ~21.76% (131.5 → 102.88 MPa) and base load by ~7.1%, with a measured weight reduction of 14.9% (≈790 kg per mast) and an estimated cost saving of ≈₹3.16 lakhs per mast (at ₹400/kg for processed fabricated steel).
- For E-ARM#2 (Ø508 mm electrode, middle arm): The redesigned mast (550 × 350 mm, similar stiffener plate) maintains an almost identical peak stress (~91 MPa for both Mast 1 and Mast 2) while reducing mast weight by ~15.4% (≈959 kg per mast, ≈₹3.84 lakhs estimated saving), confirming the cross-section reduction did not compromise structural performance.
- For E-ARM#2 (Ø457 mm electrode, middle arm): The redesigned mast (500 × 310 mm, with major-axis plate increased to 42 mm and minor-axis plate reduced to 20 mm) reduces peak stress by ~9.2%, base load by ~9.5%, and weight by ~20.4% (≈1270 kg per mast, ≈₹5.08 lakhs estimated saving) — the largest weight saving among all configurations — while total deformation remains within the allowable limit.
- For E-ARM#1 (Ø457 mm electrode, side arm): Only the original 550 × 400 mm mast was directly analyzed (peak stress 123.29 MPa, weight 4663.3 kg). Based on the favourable results already obtained for the Ø508 mm side arm and the Ø457 mm middle arm, it was concluded that the same redesign (500 × 310 mm, 42 mm / 20 mm plates) could be applied directly to this configuration to achieve similar gains, with an estimated Mast 2 weight of 3609.8 kg (≈22.6% reduction, ≈₹4.21 lakhs estimated saving), without a further dedicated optimization cycle.
- A cylindrical mast alternative, analyzed under the same loading as the Ø508 mm middle arm Mast 1, showed both higher peak stress (95.471 MPa vs. 91.017 MPa) and higher total deformation (39.791 mm vs. 32.878 mm), confirming inferior bending stiffness compared to the rectangular section; this concept was not pursued further.
- All total and directional deformation values obtained across every configuration were within the allowable deflection limits for the assembly.

12. References and Acknowledgements

12.1 Software

- Dassault Systèmes SolidWorks – 3D CAD Modelling and Mass Properties
- ANSYS Mechanical 2019 Student – Static Structural Finite Element Analysis

12.2 Standards and Material

- EN 10025 / IS 2062: Structural Steel S355 / E350
- ANSYS Mechanical APDL – Static Structural Analysis Guide

12.3 Acknowledgements

The author sincerely thanks the entire Metallurgy Department team at SMS India Pvt. Ltd. for their guidance, mentorship, and provision of design data, CAD models, and reference calculations throughout this internship.